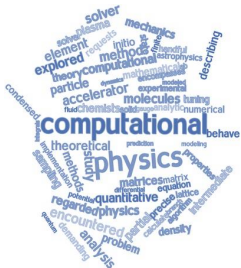


Module-2: Mathematical Physics I

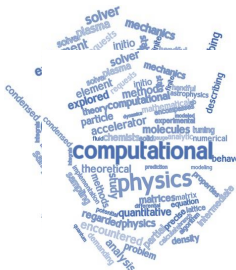
Scalars, Vector, Col linear vectors, Addition and subtraction of vectors, Scalar products, Vector products, Vector triple product, Gradient, Divergence, Curl, Matrix methods, Matrix operations, Arrays

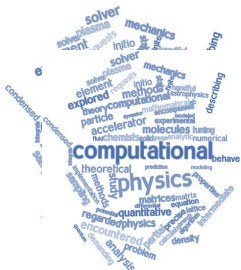
The Python logo is shown on the left. To its right is a word cloud containing terms such as "computational physics", "theoretical physics", "quantitative", "encountered", "problem", "analysis", "density", "matrix", "equation", "repeated", "physics", "polynomial", "metric", "solver", "mechanics", "describing", "element", "explored", "theory", "computation", "particle", "accelerator", "molecules", "binding", "nuclear", "statistical", "behavior", "modeling", "predictive", "modelling", "simulations", "constrained", "numerical", "hardcore", "experimental", "methodological", "approach", "into", "methods", "research", "solving", "elementary".



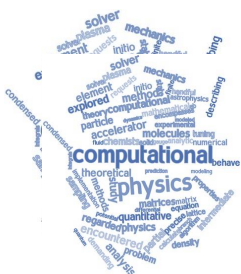
Scalars And Vectors

Scalar	Vector
Mass	Displacement
Length	Velocity
Distance	Acceleration
Speed	Force
Power	Weight
Energy	Friction
Work	Momentum





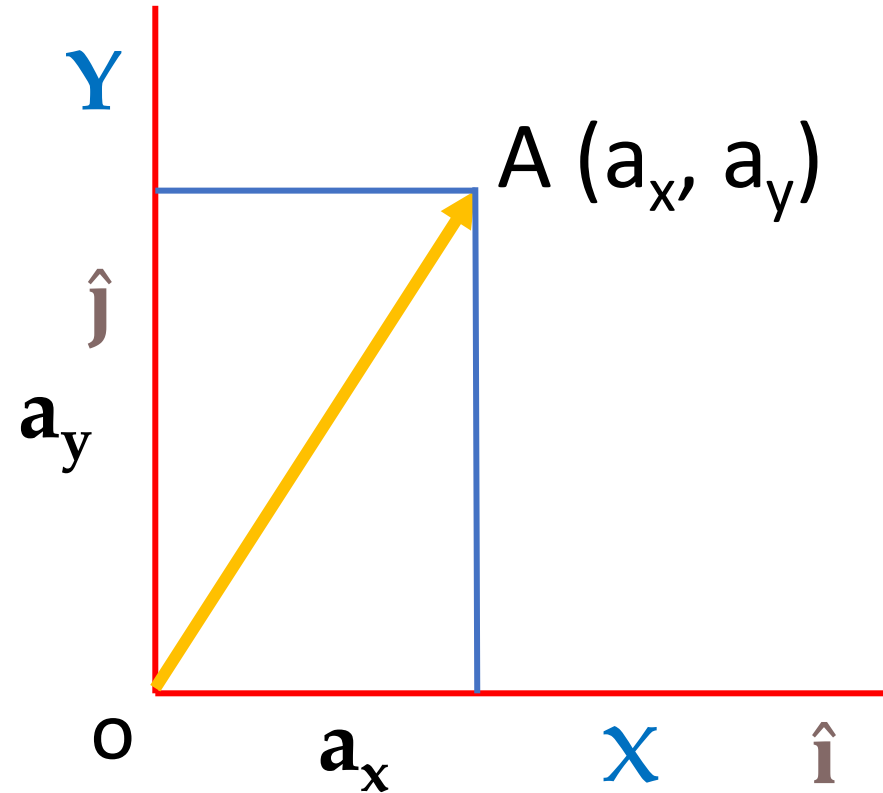




Vector

$$\overrightarrow{OA} = \vec{a} = a_x \hat{i} + a_y \hat{j} \quad [1]$$

$$\vec{a} = |\vec{a}| \hat{a} = \sqrt{a_x^2 + a_y^2} \hat{a} \quad [2]$$

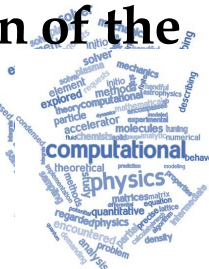


From equation [1] and [2]

$$\hat{a} = \frac{a_x \hat{i} + a_y \hat{j}}{\sqrt{a_x^2 + a_y^2}}$$

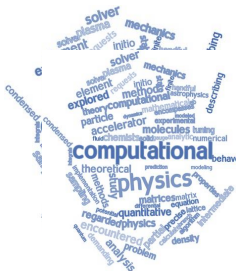
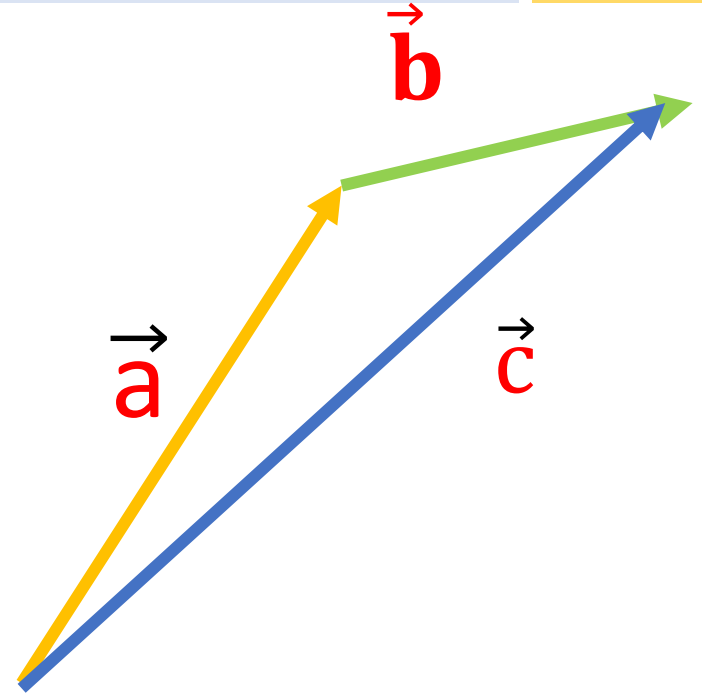
$$\vec{a} = \underbrace{(\sqrt{a_x^2 + a_y^2})}_{\text{Magnitude of the vector } \vec{a}} \frac{a_x \hat{i} + a_y \hat{j}}{\sqrt{a_x^2 + a_y^2}}$$

Direction of the vector $\vec{a}$



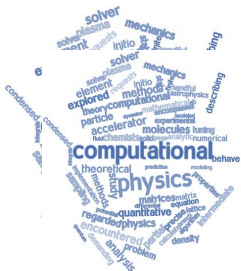
Vector Addition ($\vec{a} + \vec{b}$)

$$\vec{c} = \vec{a} + \vec{b}$$





A 2D Cartesian coordinate system with a horizontal axis labeled $\hat{i}$ and a vertical axis labeled $\hat{j}$. The origin is marked with a bold black 'O'. Three vectors originate from the origin: a yellow vector labeled $\vec{a}$ pointing to point A, a green vector labeled $\vec{b}$ pointing to point B, and a blue vector labeled $\vec{c}$ pointing to point B. The vector $\vec{c}$ is the resultant of $\vec{a}$ and $\vec{b}$, as indicated by the triangle rule where $\vec{a}$ and $\vec{b}$ are connected end-to-end to form $\vec{c}$.



Vector Addition ($\vec{a} + \vec{b}$)

$$\vec{b} = b_x \hat{i} + b_y \hat{j}$$

$$= (c_x - a_x) \hat{i} + (c_y - a_y) \hat{j}$$

$$= \underbrace{(c_x \hat{i} + c_y \hat{j})}_{\vec{c}} - \underbrace{(a_x \hat{i} + a_y \hat{j})}_{\vec{a}}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} = \vec{c} - \vec{a}$$

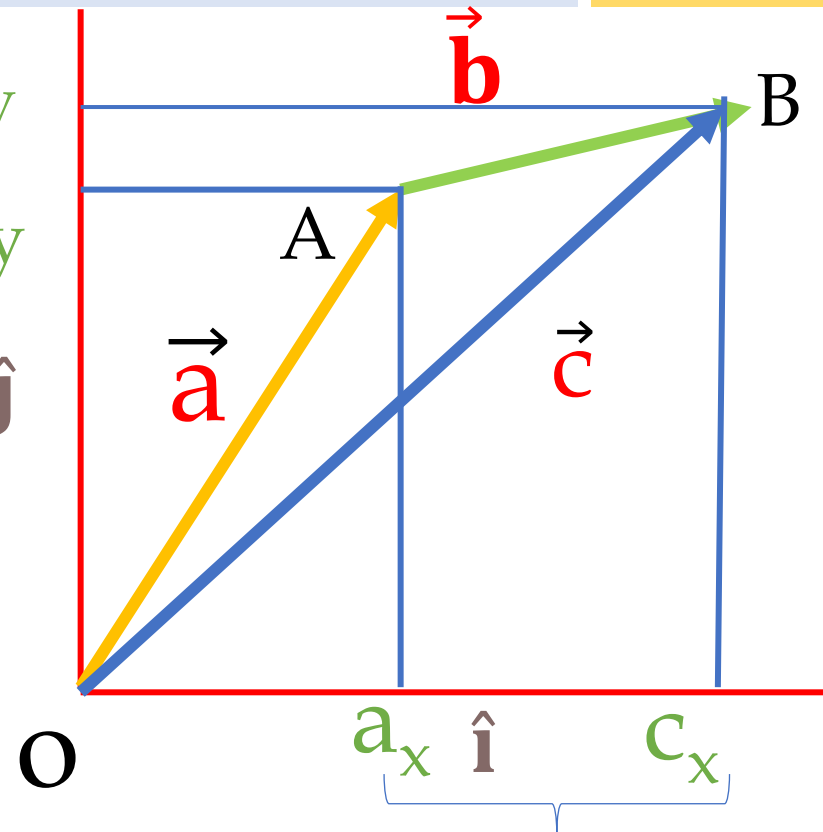
$$\vec{b} = \vec{c} - \vec{a}$$

$$\vec{c} = \vec{a} + \vec{b}$$

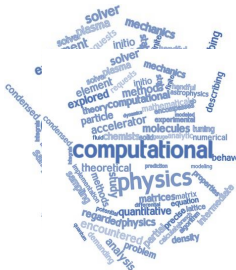
$$b_y = c_y - a_y$$

c_y
 a_y

$\hat{j}$

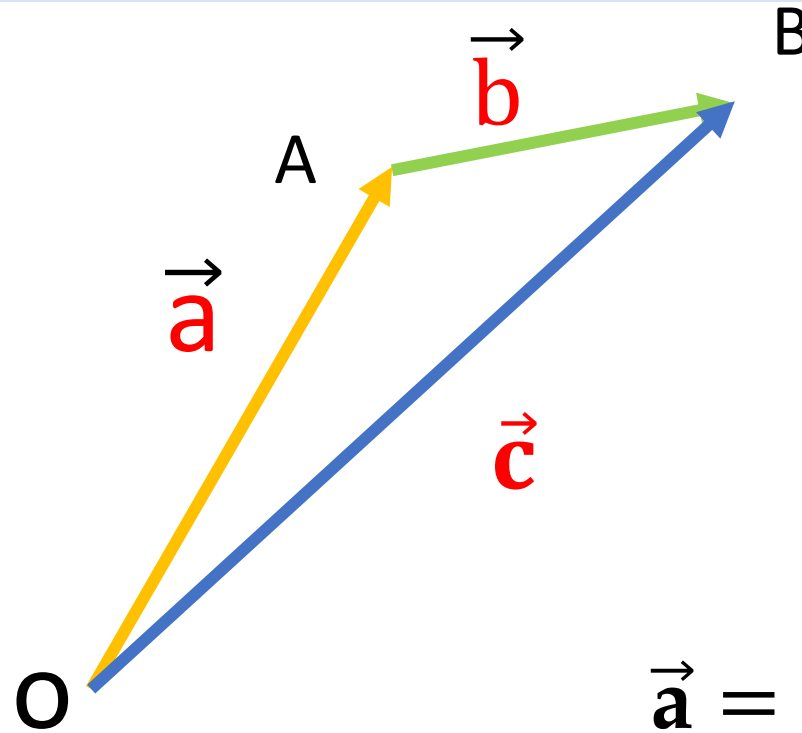


$$b_x = c_x - a_x$$



Vector Addition ($\vec{a} + \vec{b}$)

$$\vec{c} = \vec{a} + \vec{b}$$

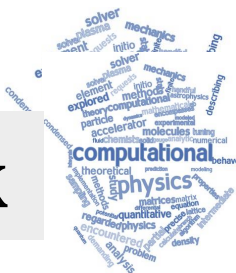


$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$\vec{c} = \vec{a} + \vec{b}$$

$$\vec{c} = (a_x + b_x) \hat{i} + (a_y + b_y) \hat{j} + (a_z + b_z) \hat{k}$$



The diagram shows a vector $\vec{a}$ (yellow) starting at point O and ending at point A. A second vector $\vec{b}$ (green) starts at point A and ends at point B. A third vector $\vec{c}$ (blue) starts at point O and ends at point B. The vectors are labeled with red text and arrows. The equation $\vec{a} = \vec{c} - \vec{b}$ is written in red at the bottom right.

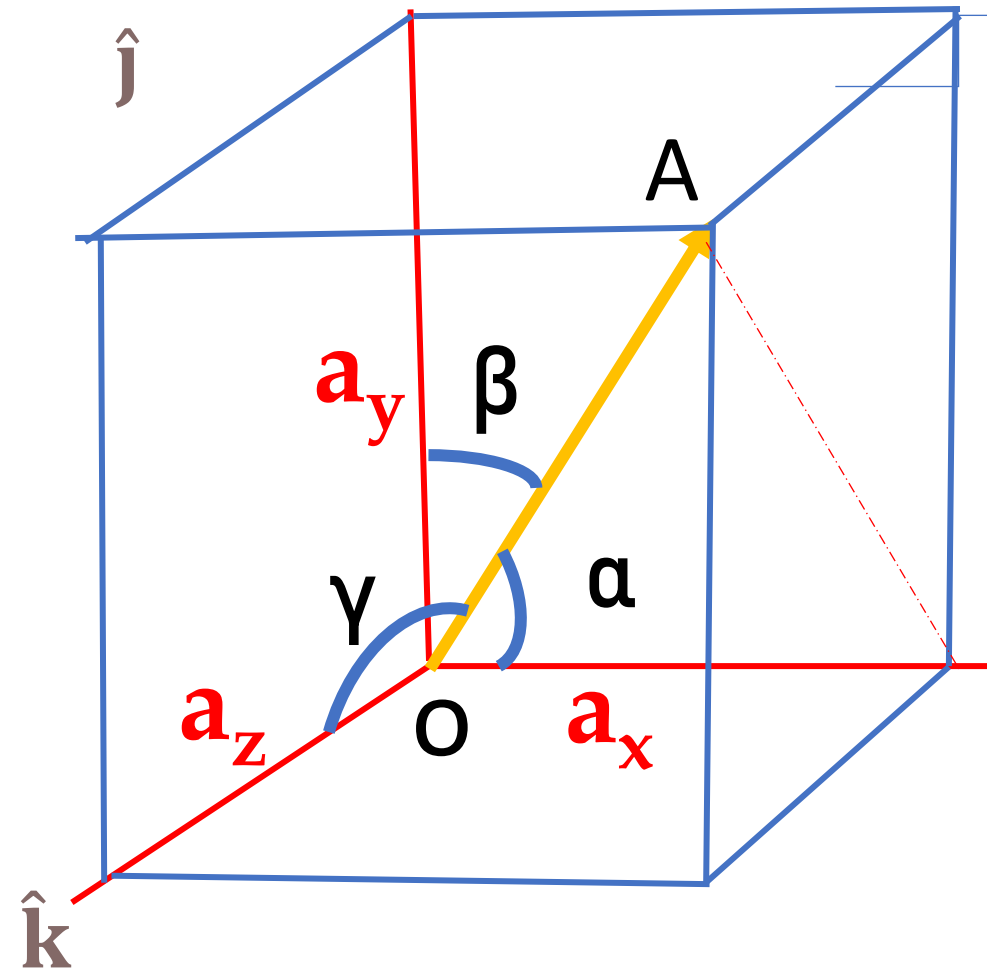
$$\vec{\mathbf{b}} = \mathbf{b}_x \hat{\mathbf{i}} + \mathbf{b}_y \hat{\mathbf{j}} + \mathbf{b}_z \hat{\mathbf{k}}$$





$$\frac{\vec{a}}{|\vec{a}|} = \hat{a}$$

Vector in 3D



Cuboid holding vector $\overrightarrow{OA} = \vec{a}$ along the diagonal

$$\overrightarrow{OA} = \vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\cos(\alpha) = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{\vec{a} \cdot \hat{i}}{OA} = \frac{a_x}{\sqrt{a_x^2 + a_y^2 + a_z^2}} \quad [1]$$

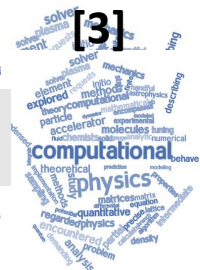
$$\cos(\beta) = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{\vec{a} \cdot \hat{j}}{OA} = \frac{a_y}{\sqrt{a_x^2 + a_y^2 + a_z^2}} \quad [2]$$

$$\cos(\gamma) = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{\vec{a} \cdot \hat{k}}{OA} = \frac{a_z}{\sqrt{a_x^2 + a_y^2 + a_z^2}} \quad [3]$$

Adding square of cosine angles from equations[1], [2] and [3]

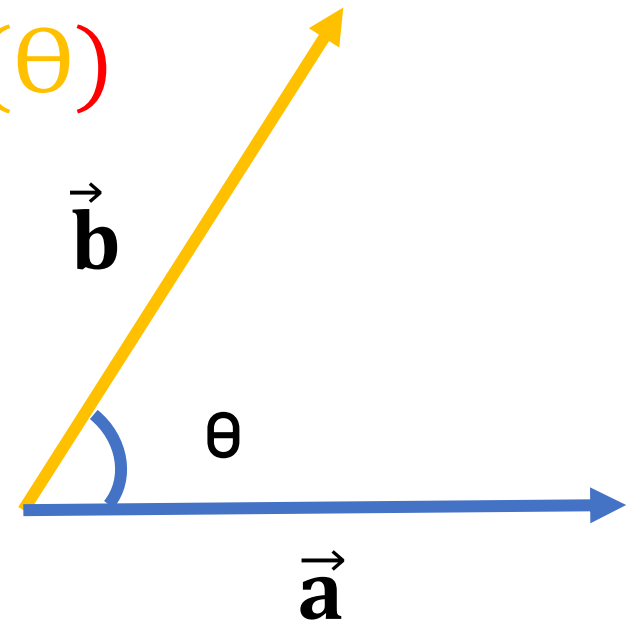
$$\cos^2(\alpha) + \cos^2(\beta) + \cos^2(\gamma) = 1$$

<https://www.youtube.com/watch?v=Uws4tmZbNtU>

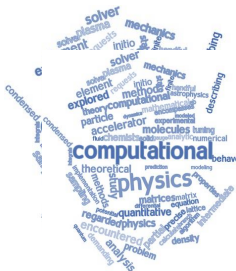


Dot Product ($\vec{a} \cdot \vec{b}$)

$$c = \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos(\theta)$$

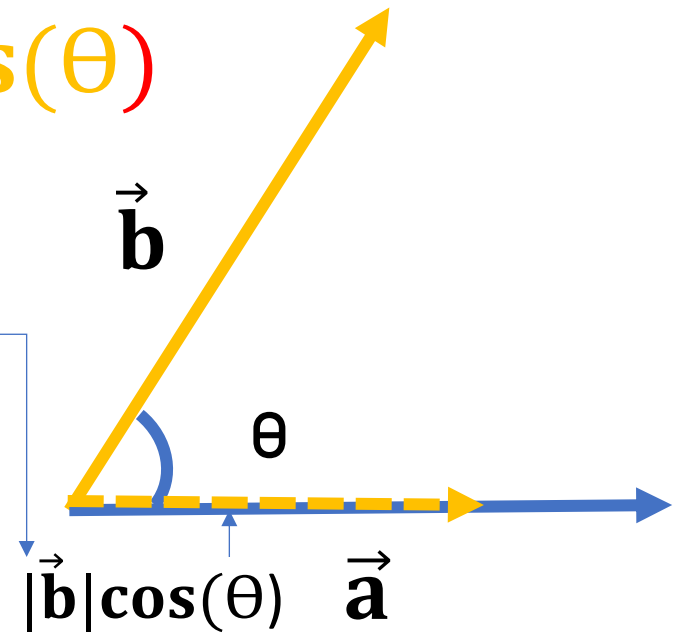


This product gives a scalar value and represents the projection of one vector onto another.



Vector Dot Product ($\vec{a} \cdot \vec{b}$)

$$c = \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos(\theta)$$



$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$c = \vec{a} \cdot \vec{b}$$



$$c = a_x b_x + a_y b_y + a_z b_z$$

The dot product of $\mathbf{u} = \langle u_1, u_2 \rangle$ and $\mathbf{v} = \langle v_1, v_2 \rangle$ is given by $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2$

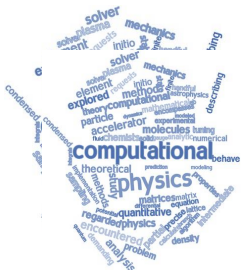
Properties of the Dot Product

Let $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be vectors and let c be a scalar.

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $\mathbf{0} \cdot \mathbf{v} = 0$
3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
4. $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$
5. $c(\mathbf{u} \cdot \mathbf{v}) = c\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot c\mathbf{v}$

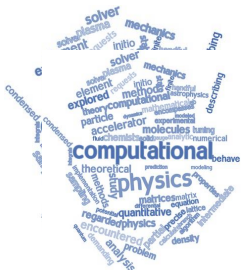
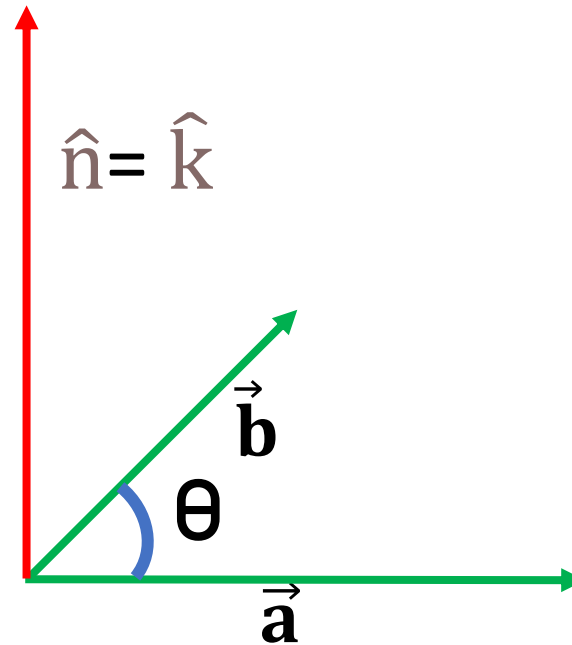


$$\vec{\mathbf{b}} = -8\hat{\mathbf{i}} - 12\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$$



$$\vec{A} = \vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin(\theta) \hat{n}$$

The vector product (cross product) $\vec{A}$ of two vectors $\vec{a}$ and $\vec{b}$ gives a new vector whose magnitude represents the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$, with the direction perpendicular to the plane containing them



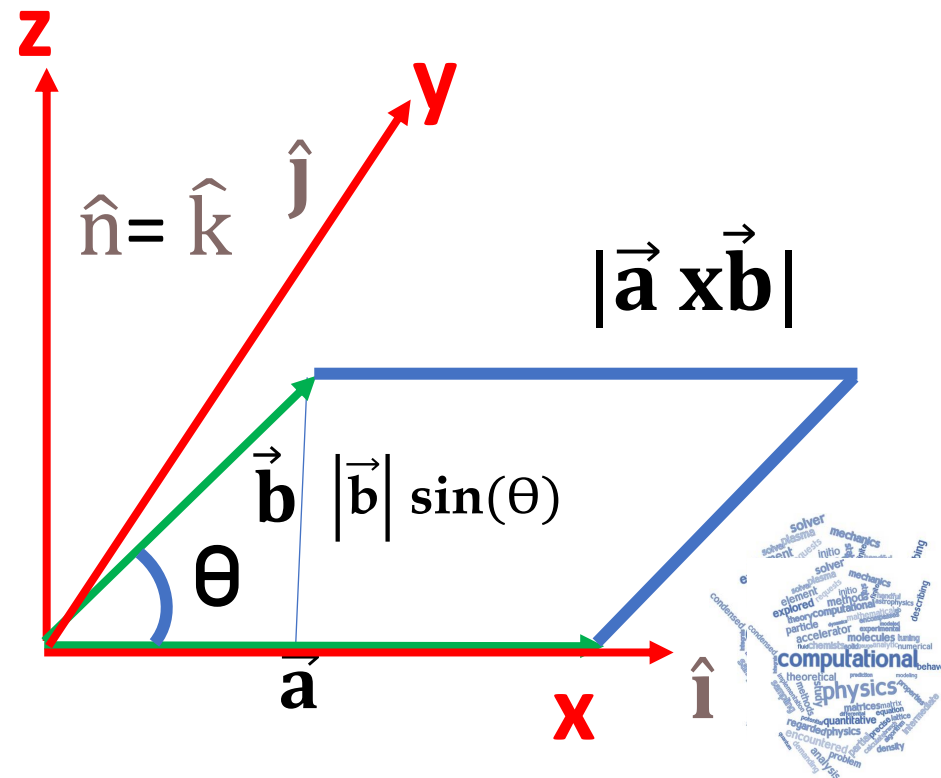
Area (A) of Parallelogram = Base X Perpendicular Distance

$$\text{Area (A)} = |\vec{a}| |\vec{b}| \sin(\theta)$$

$$\vec{A} = |\vec{a}| |\vec{b}| \sin(\theta) \hat{n}$$

$$\vec{A} = \vec{a} \times \vec{b}$$

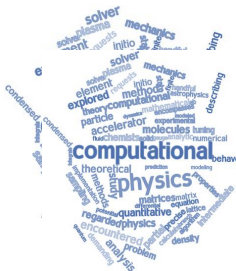
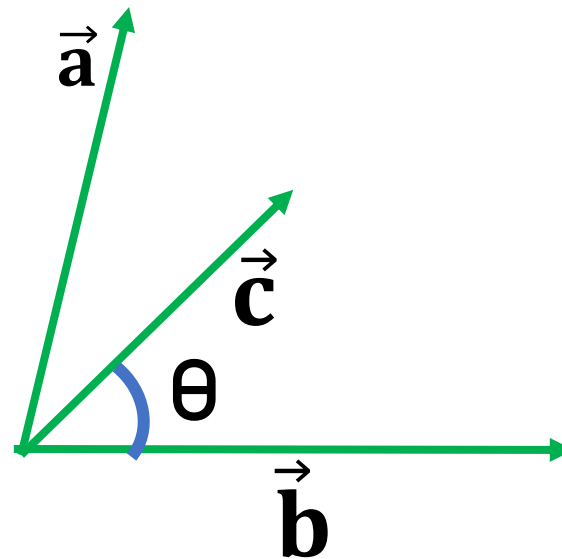
**Area of the parallelogram
formed by $\vec{a}$ and $\vec{b}$.**



Scalar Triple Product($\vec{a} \cdot (\vec{b} \times \vec{c})$)

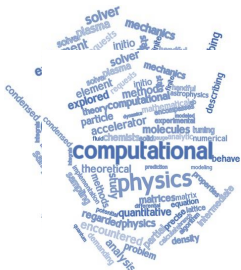
$$\text{Scalar triple product } V = \vec{a} \cdot (\vec{b} \times \vec{c})$$

The scalar triple product of three vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ gives the volume of the parallelepiped formed by these vectors.



Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$, and $\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$

then
$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \\ \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \\ \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 \end{bmatrix}$$



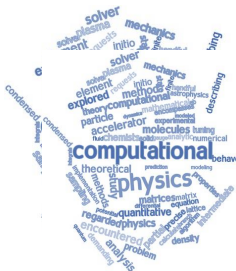
Question ?

$$\vec{a} = -4\hat{i} - 6\hat{j} - 2\hat{k}$$

$$\vec{b} = -\hat{i} + 4\hat{j} + 3\hat{k}$$

$$\vec{c} = -8\hat{i} - \hat{j} + 3\hat{k}$$

Show $\vec{a}$, $\vec{b}$, and $\vec{c}$ are the co-planer i.e. $[\vec{a} \ \vec{b} \ \vec{c}] = 0$



Vector Triple Product

If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are three vector quantities, then $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ represents the vector triple product and is given by

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{b} \cdot \mathbf{c}) \mathbf{a}$$
$$\Rightarrow \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$$

Vector Triple Product

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Vector Triple Product

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$$\Rightarrow \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$$

Vector Triple Product

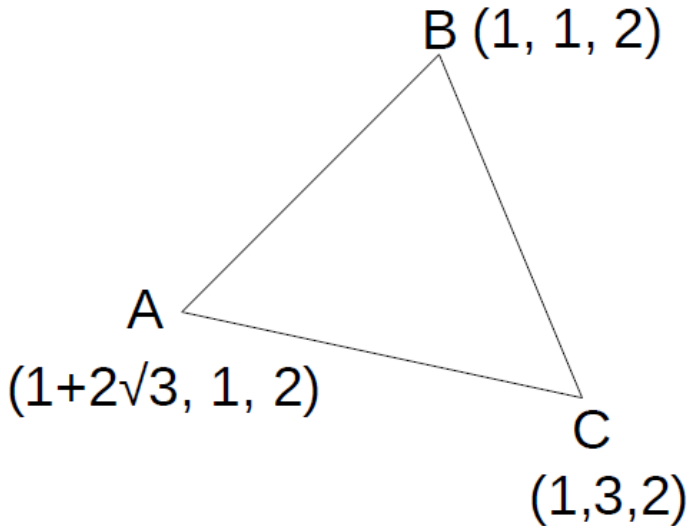
If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are three vector quantities, then $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ represents the vector triple product and is given by

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{b} \cdot \mathbf{c}) \mathbf{a}$$
$$\Rightarrow \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$$

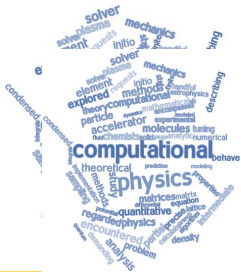
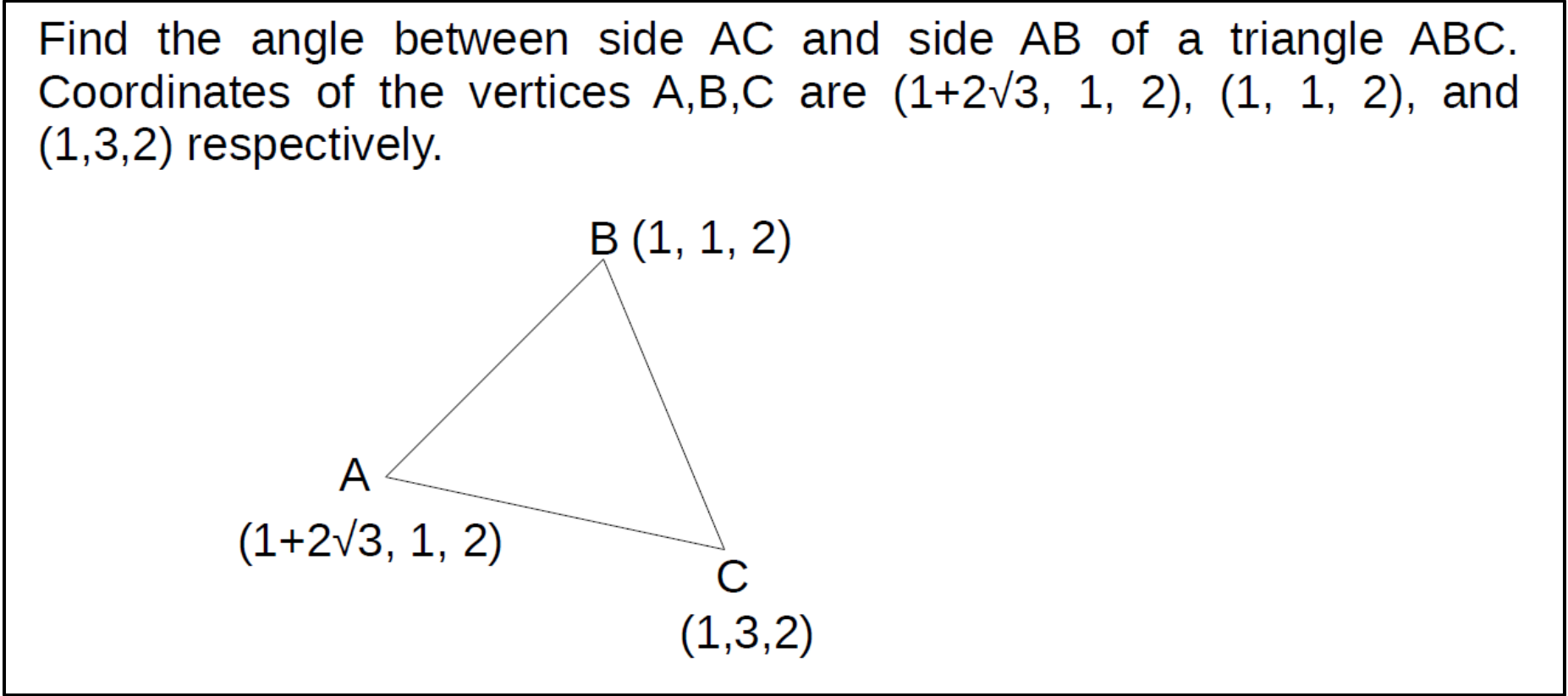
If **a**, **b**, and **c** are vectors and c is a scalar, then

1. $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$
2. $(c\mathbf{a}) \times \mathbf{b} = c(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (c\mathbf{b})$
3. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
4. $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$
5. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$
6. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$

Find the angle between side AC and side AB of a triangle ABC. Coordinates of the vertices A,B,C are $(1+2\sqrt{3}, 1, 2)$, $(1, 1, 2)$, and $(1,3,2)$ respectively.

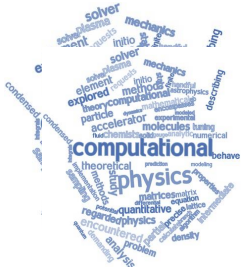


The diagram shows a triangle ABC in a 3D coordinate system. The vertices are labeled with their coordinates: A is at $(1+2\sqrt{3}, 1, 2)$, B is at $(1, 1, 2)$, and C is at $(1, 3, 2)$. The triangle is formed by connecting these three points.



Write a program to perform scalar and vector multiplication, vector addition and subtraction, and vector dot and cross products using list and for loop?

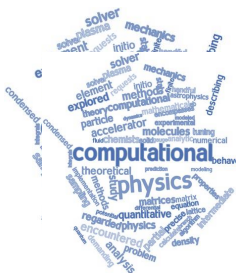
Look at the Jupyter Notebook for this !!



Write a program to perform scalar and vector multiplication, vector addition and subtraction, and vector dot using 1D-Array

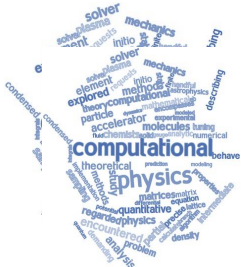
?

Look at the Jupyter Notebook for this !!



Write a code for Vector dot product and cross product using 1D Array and cross and dot function from library ?

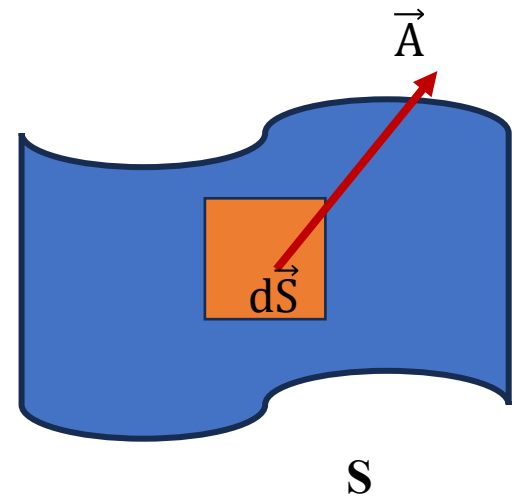
Look at the Jupyter Notebook for this !!



Let us know some basic things

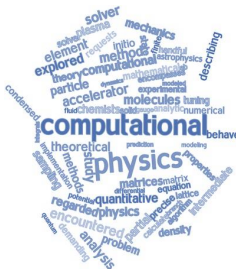
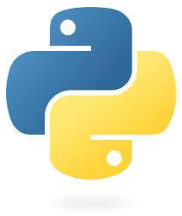
Surface Integral

A surface integral is an integral in which the function to be integrated over the surface S



The Surface integral of the vector $\vec{A}$ $= \int_S \vec{A} \cdot d\vec{S}$

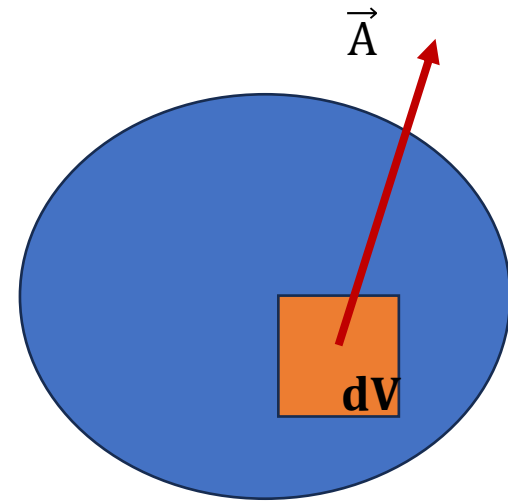
Vector flux passing through the surface



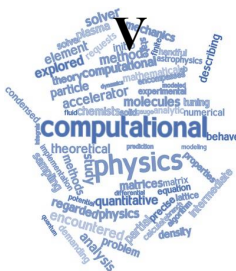
Let us know Some Basic Things

Volume Integral

A volume integral is an integral in which the function to be integrated over the volume V



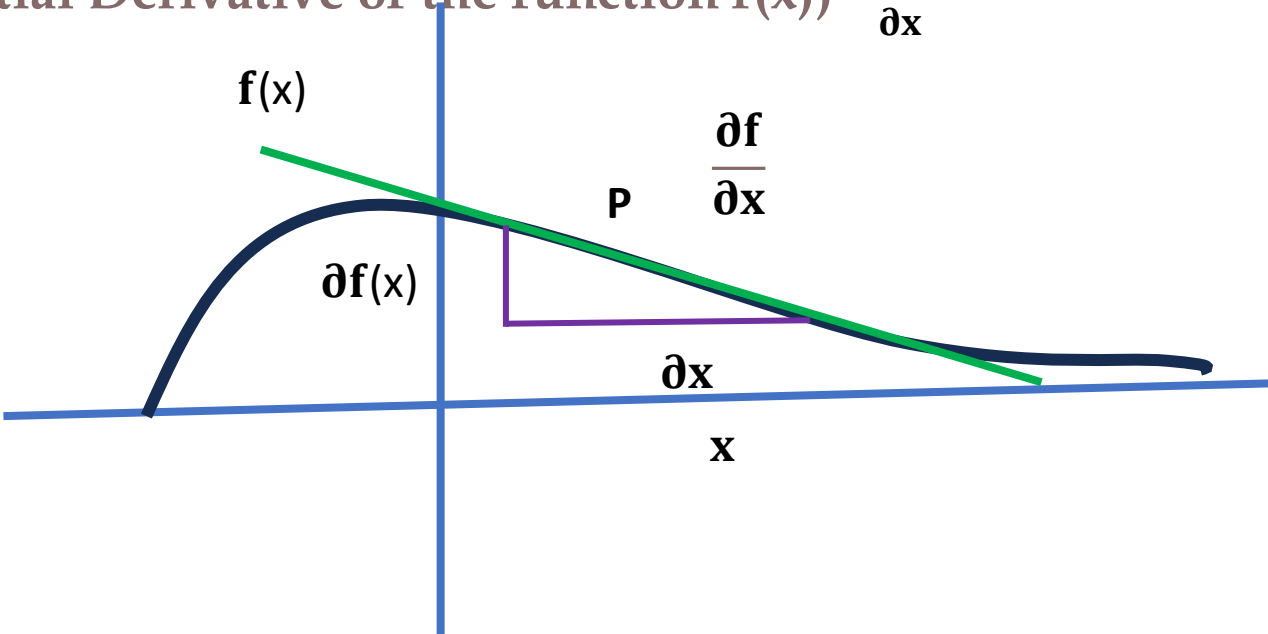
The Volume integral of the vector $\vec{A}$ $= \int_V \vec{A} dV$



Derivative

$$\frac{\partial}{\partial \mathbf{x}}$$

Partial Derivative of the function $f(\mathbf{x}) = \frac{\partial f}{\partial \mathbf{x}}$

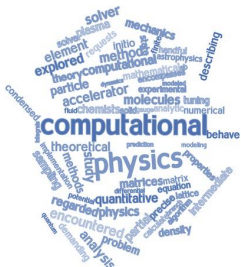
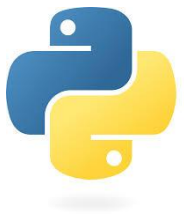


is the rate of the change of the function f with x .



$$\vec{\nabla} = \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z}$$

- The del operator is the differential operator which is represented by $\vec{\nabla}$
- It is not a vector itself but when it operates on a function it provides the resultant as a vector or scalar.



Gradient

Let $\phi(x, y, z)$ be the scalar field or scalar function. Then gradient of the scalar function can be defined as

$$\text{Grad } \phi = \vec{\nabla} \phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \phi$$

$$= \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

where $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$ is called the Del operator

- $\text{Grad } \phi = \vec{\nabla} \phi$ is the vector quantity.
- The gradient of a scalar quantity signifies the direction of the greatest change of the quantity.

Change in the function $\phi(x, y, z)$ can be written as

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$=(\hat{\mathbf{i}}\frac{\partial\Phi}{\partial x}+\hat{\mathbf{j}}\frac{\partial\Phi}{\partial y}+\hat{\mathbf{k}}\frac{\partial\Phi}{\partial z}).(\mathbf{d}x\hat{\mathbf{i}}+\mathbf{d}y\hat{\mathbf{j}}+\mathbf{d}z\hat{\mathbf{k}})$$

$$\vec{\nabla} \Phi$$

$$d\vec{r}$$

$$d\phi = \vec{\nabla} \phi \cdot d\vec{r}$$

$$d\phi = |\vec{\nabla}\phi| \, |d\vec{r}| \cos\theta$$

where θ is the angle between $\vec{\nabla}\phi$ and $d\vec{r}$

Example: The **temperature function** $\Phi(x, y, z) = T(x, y, z)$ describes the temperature distribution in a region of space. If derived from a **heat element** and placed at **O**



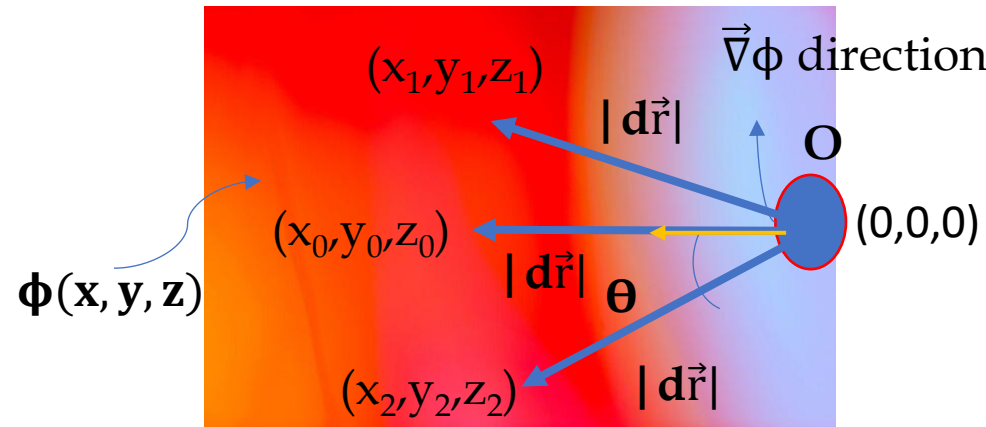
Gradient: Physical Interpretation

Thus, change in the function $d\phi$ can be written as

$$d\phi = d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$d\phi = \vec{\nabla} \phi \cdot d\vec{r}$$

$$d\phi_{\max} = |\vec{\nabla} \phi| |d\vec{r}| \cos(0)$$



Maximum temperature change $d\phi_{\max}$ along the direction of gradient of the function ϕ

Magnitude $|\vec{\nabla} \phi|$ this tells us how rapidly the function ϕ is changing at that point. The larger the magnitude, the steeper the change

Direction: The gradient points in the direction of the fastest increase of ϕ

The gradient of a scalar function ϕ ($\nabla \phi$) is a vector whose magnitude represents the rate of maximum change in the function with respect to the spatial variables, and its direction corresponds to the direction of the steepest ascent of the function.

Gradient: Question?

Prove $\text{Grad} \frac{1}{|\vec{r}|^n} = \frac{-n\vec{r}}{|\vec{r}|^{n+2}}$ where $\vec{r}$ is the position vector.

$$\text{Grad} \frac{1}{|\vec{r}|^n} = \vec{\nabla} \frac{1}{|\vec{r}|^n} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \frac{1}{|\vec{r}|^n}$$

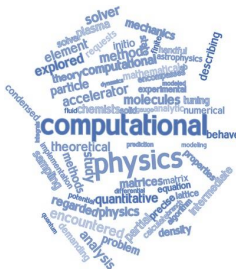
Where $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \frac{1}{(\sqrt{x^2 + y^2 + z^2})^n}$$

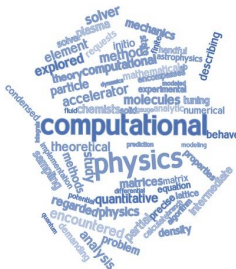
$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \frac{1}{(x^2 + y^2 + z^2)^{n/2}}$$

$$\text{Grad} \frac{1}{|\vec{r}|^n} = \vec{\nabla} \frac{1}{|\vec{r}|^n} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \frac{1}{(x^2 + y^2 + z^2)^{n/2}}$$



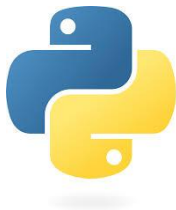
Question?

$$\begin{aligned}\text{Grad} \frac{1}{|\vec{r}|^n} &= \vec{\nabla} \frac{1}{|\vec{r}|^n} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \frac{1}{(x^2 + y^2 + z^2)^{n/2}} \\&= \hat{i} \frac{\partial}{\partial x} \frac{1}{(x^2 + y^2 + z^2)^{n/2}} + \hat{j} \frac{\partial}{\partial y} \frac{1}{(x^2 + y^2 + z^2)^{n/2}} + \hat{k} \frac{\partial}{\partial z} \frac{1}{(x^2 + y^2 + z^2)^{n/2}} \\&= \frac{-n}{2(x^2 + y^2 + z^2)^{\frac{n+2}{2}}} 2x\hat{i} + \frac{-n}{2(x^2 + y^2 + z^2)^{\frac{n+2}{2}}} 2y\hat{j} + \frac{-n}{2(x^2 + y^2 + z^2)^{\frac{n+2}{2}}} 2z\hat{k} \\&= \frac{-n}{|\vec{r}|^{n+2}} x\hat{i} + \frac{-n}{|\vec{r}|^{n+2}} y\hat{j} + \frac{-n}{|\vec{r}|^{n+2}} z\hat{k} \\&= \frac{-n}{|\vec{r}|^{n+2}} (x\hat{i} + y\hat{j} + z\hat{k}) \\&= \frac{-n\vec{r}}{|\vec{r}|^{n+2}} \\ \text{Grad} \frac{1}{|\vec{r}|^n} &= \frac{-n\vec{r}}{|\vec{r}|^{n+2}}\end{aligned}$$



Consider the scalar function $f(x,y)=x^2+3y$.

1. Calculate the gradient of the function $f(x,y)$, denoted by $\vec{\nabla}f$
2. Evaluate the $\vec{\nabla}f$ at the $P(2,-1)$
3. Determine the direction of maximum change of f at point $(2,-1)$

- Direction of the maximum change is**
- i. e. $\frac{\vec{\nabla} f}{|\vec{\nabla} f|}$**
- 
- $$\frac{\vec{\nabla} f}{|\vec{\nabla} f|} =$$

$$\vec{\nabla}f \text{ at the point P } (2, -1) = 4\hat{i} + 3\hat{j}$$

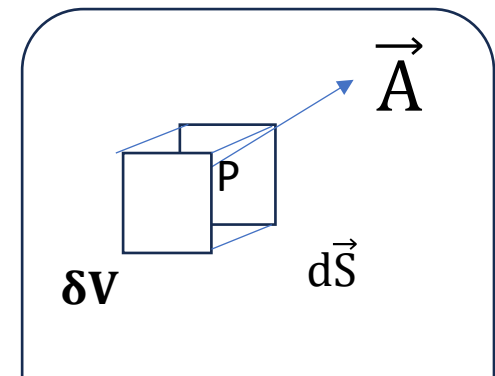
i. e. $\frac{\nabla f}{\|\nabla f\|}$

Divergence of the Vector

Let there exist a vector field $\vec{A}$ in a certain region of space, the divergence at the certain point $P(x,y,z)$ is defined as the outwards flux of vector field $\vec{A}$ per unit volume enclosed, through an infinitesimal closed surface surrounded the point P .

Let δV is an infinitesimal volume enclosed by an infinitely small closed-surfaced $d\vec{S}$ surrounded the point $P(x,y,z)$

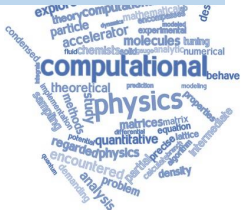
$$\text{div} \vec{A} = \vec{\nabla} \cdot \vec{A} = \lim_{\delta V \rightarrow 0} \frac{\iint \vec{A} \cdot d\vec{S}}{\delta V} = \frac{\text{Vector Flux}}{\text{Volume}}$$



Physically, we may regard the divergence of a vector field $\vec{A}$ at a given point as a measure of how much the field diverges from that point.

The name divergence is well-chosen as it measures of much the vector spreads out (diverges) from the reference point

$$\text{div} \vec{A} = \vec{\nabla} \cdot \vec{A} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (\hat{i} A_x + \hat{j} A_y + \hat{k} A_z) = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$



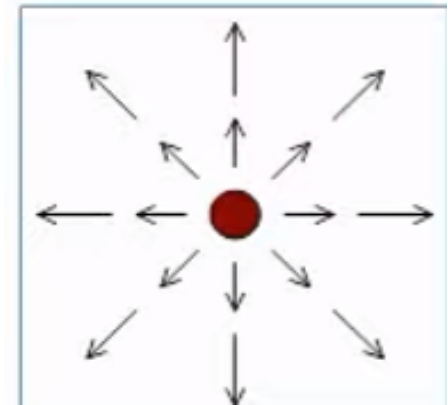
Divergence of the Vector

$$\vec{A} = x\hat{i} + y\hat{j} + z\hat{k}$$

[This vector function has a large positive divergence].

$$\vec{\nabla} \cdot \vec{A} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x\hat{i} + y\hat{j} + z\hat{k})$$

$$= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3$$

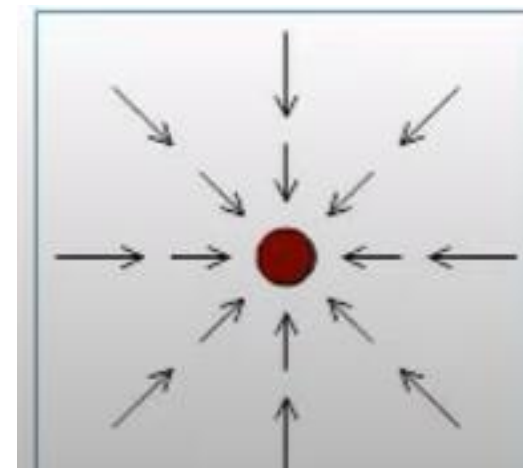


$$\vec{A} = -x\hat{i} - y\hat{j} - z\hat{k}$$

[This vector function has a large negative divergence].

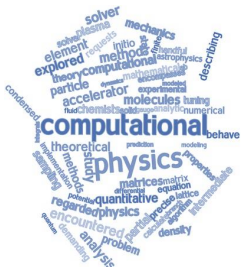
$$\vec{\nabla} \cdot \vec{A} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (-x\hat{i} - y\hat{j} - z\hat{k})$$

$$= -\frac{\partial x}{\partial x} - \frac{\partial y}{\partial y} - \frac{\partial z}{\partial z} = -1 - 1 - 1 = -3$$



A diagram consisting of a rectangular box containing three vertical columns of arrows. Each column contains three arrows pointing to the left, for a total of nine arrows.

$$\vec{A} = y^2 \hat{i} + (2xy + z^2) \hat{j} - 2yz \hat{k}, \text{ find } \vec{\nabla} \cdot \vec{A}$$



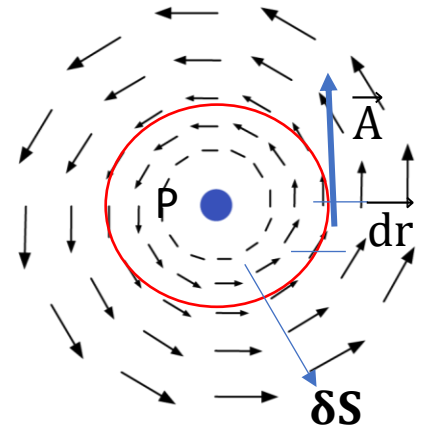
Curl of the Vector

A curl of a vector field $\vec{A}$ at any point is defined as a vector quantity whose magnitude is equal to the maximum line integral per unit area along the boundary of an infinitesimal test area δS at the point and whose direction is perpendicular to the plane of the test area.

$$|\text{Curl} \vec{A}| = \lim_{\delta S \rightarrow 0} \frac{|\int_c \vec{A} \cdot d\vec{r}|_{\max}}{\delta S} = \frac{\text{Maximum line integral of Vector } \vec{A}}{\text{Volume}}$$

$$\text{Curl} \vec{A} = \lim_{\delta S \rightarrow 0} \frac{|\int_c \vec{A} \cdot d\vec{r}|_{\max}}{\delta S} \hat{n}$$

Where, $\hat{n}$ is the unit vector along the normal to the plane of the test area δS



The curl of the vector field $\vec{A}$ is sometimes defined as circulation.

The name curl is also well chosen as it measures of how much the vector curls around the point.

Curl of the Vector

When the operator $\vec{\nabla}$ acts on a vector function via cross product, we get the curl of a vector function $\vec{A}$,

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

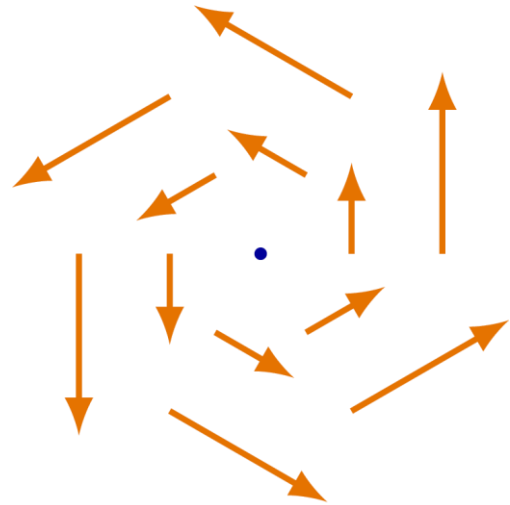
$$\text{Curl } \vec{A} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\text{Curl } \vec{A} = \vec{\nabla} \times \vec{A} = \hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{j} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

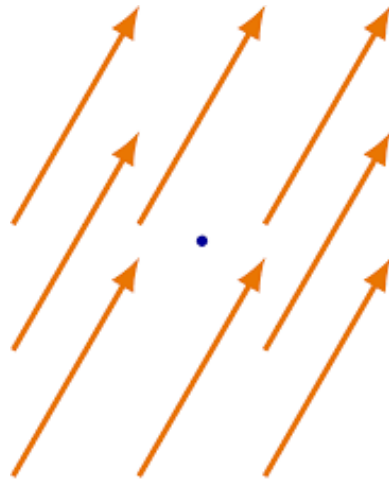
$\vec{\nabla} \times \vec{A}$ is a measure of how much the vector $\vec{A}$ curls around the given point. Zero curl means there is no rotation.

The curl of a vector function is a vector.

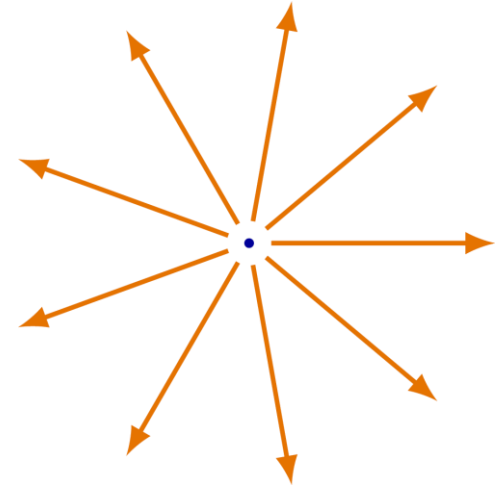
Curl of the Vector



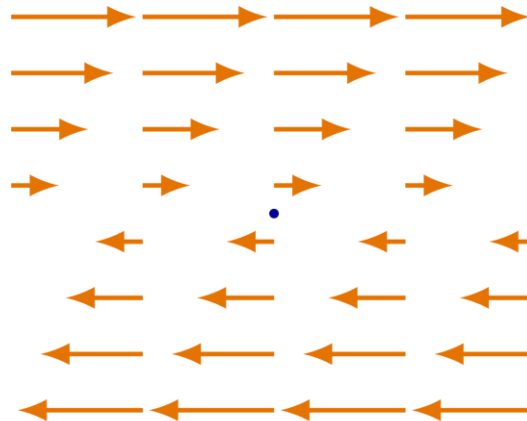
$$\nabla \times \mathbf{v} = \odot$$



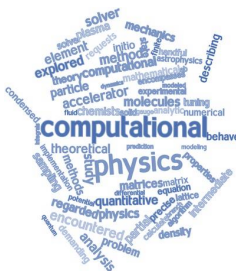
$$\nabla \times \mathbf{v} = 0$$



$$\nabla \times \mathbf{v} = 0$$



$$\nabla \times \mathbf{v} = \otimes$$



Prove that electric field $\vec{E}$ generated by point charge Q is the irrotational i.e. $\text{Curl } \vec{E} = \vec{\nabla} \times \vec{E} = 0$

Hint

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} = \frac{Q}{4\pi\epsilon_0 r^3} \vec{r}$$

$$\vec{r} = (x\hat{i} + y\hat{j} + z\hat{k})$$

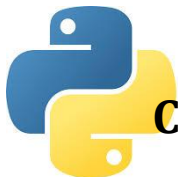
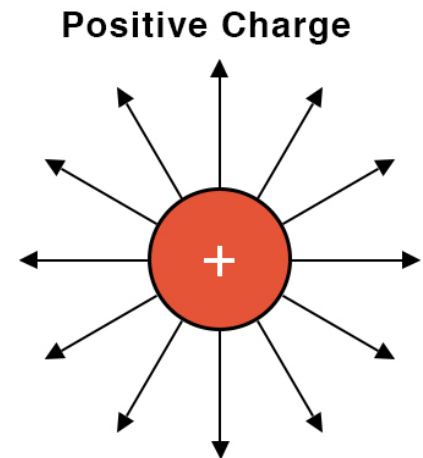
$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\mathbf{r}^3 = (\mathbf{x}^2 + \mathbf{y}^2 + \mathbf{r}^2)^{3/2}$$

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} = \frac{Q}{4\pi\epsilon_0 r^3} \vec{r}$$

$$\vec{E} = \frac{Q}{4\pi\epsilon_0(x^2 + y^2 + z^2)^{3/2}} (x\hat{i} + y\hat{j} + z\hat{k})$$

$$\text{Curl } \vec{E} = \vec{\nabla} \times \vec{E} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times \frac{Q}{4\pi\epsilon_0(x^2+y^2+z^2)^{3/2}} (\hat{x} + \hat{y} + \hat{z}) = ? = 0$$

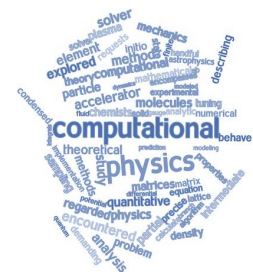


If vector function $\bar{A} = -iy+jx$, Calculate the its curl

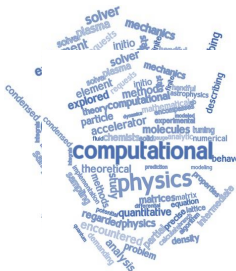
$$\text{Curl}\vec{A} = \vec{\nabla} \times \vec{A} = (\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}) \times (-y\hat{i} + x\hat{j})$$

$$\text{Curl}\vec{A} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y & x & 0 \end{vmatrix}$$

$$\text{Curl}\vec{A} = \vec{\nabla} \times \vec{A} = \hat{i}\left(\frac{\partial 0}{\partial y} - \frac{\partial x}{\partial z}\right) + \hat{j}\left(\frac{\partial(-y)}{\partial z} - \frac{\partial 0}{\partial x}\right) + \hat{k}\left(\frac{\partial x}{\partial x} - \frac{\partial(-y)}{\partial y}\right) = i0 + j0 + k(1 - (-1)) = 2\hat{k}$$



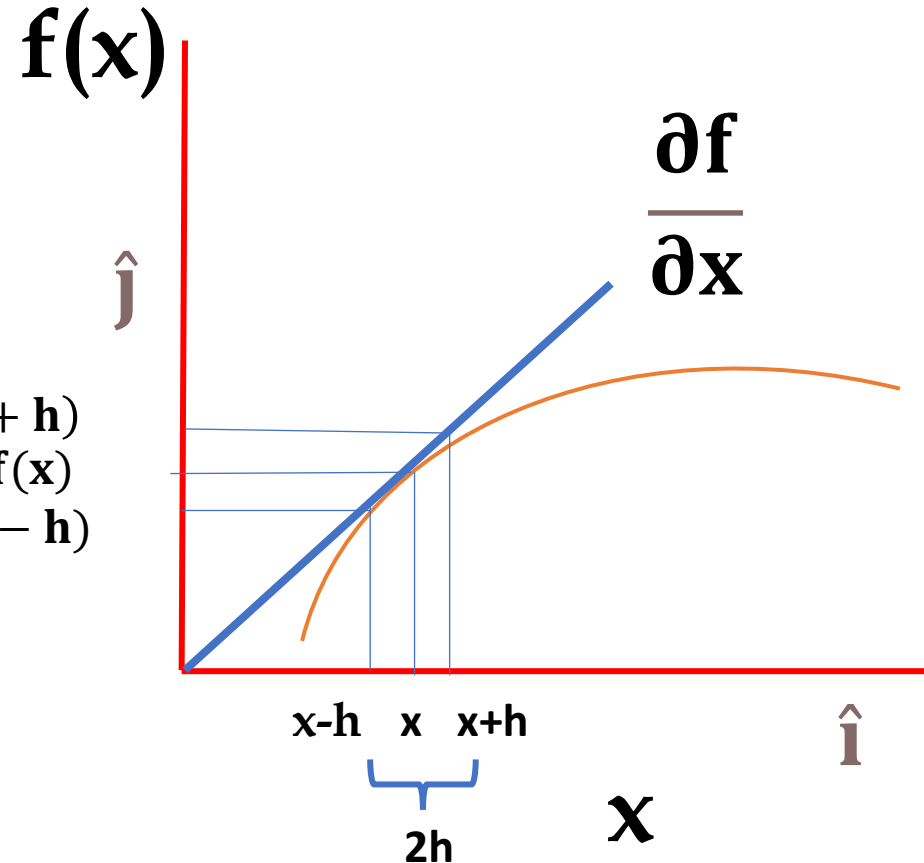
**Look at the
Jupyter Notebook
for this !!**



Partial Differentiation $\frac{\partial f}{\partial x}$ using Numerical Method

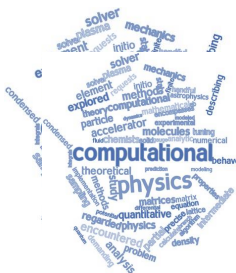
Central Difference Method

$$f(x+h) - f(x-h) \quad \left\{ \begin{array}{l} f(x+h) \\ f(x) \\ f(x-h) \end{array} \right.$$



$$\frac{\partial f}{\partial x} \approx \frac{f(x+h) - f(x-h)}{2h}$$

h is the small size

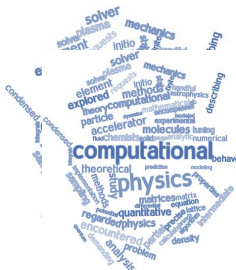


Let us check the partial derivative of a function $f(x) = \sin(x)$ using the numerical method:

$$\frac{\partial f(x)}{\partial x} = \cos x$$

$$\frac{\partial f(x)}{\partial x} = 0 \quad \text{at } x = 90^\circ$$

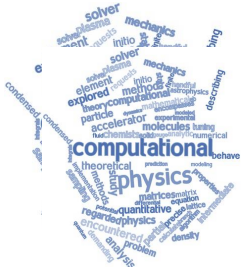
Look at the Jupyter Notebook for this !!



$$\vec{\nabla} f(x, y, z) = \hat{i} \frac{f(x+h, y, z) - f(x-h, y, z)}{2h} + \hat{j} \frac{f(x, y+h, z) - f(x, y-h, z)}{2h} + \hat{k} \frac{f(x, y, z+h) - f(x, y, z-h)}{2h}$$

Write a code for determining gradient of scalar function $f(x,y,z)$ at the given point.

Look at the Jupyter Notebook for this !!

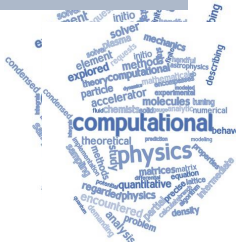


The Divergence of $\vec{A}(x, y, z)$ using Numerical Method

$$\frac{\partial f}{\partial x} \approx \frac{f(x+h) - f(x-h)}{2h}$$

$$\begin{aligned}\text{div} \vec{A} = \vec{\nabla} \cdot \vec{A} &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (\hat{i} A_x + \hat{j} A_y + \hat{k} A_z) \\ &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\end{aligned}$$

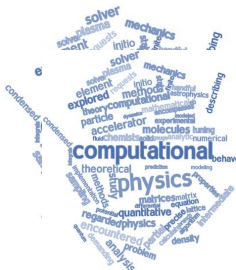
$$\vec{\nabla} \cdot \vec{A} = \frac{A_x(x+h, y, z) - A_x(x-h, y, z)}{2h} + \frac{A_y(x, y+h, z) - A_y(x, y-h, z)}{2h} + \frac{A_z(x, y, z+h) - A_z(x, y, z-h)}{2h}$$



The Divergence of $\vec{A}(x, y, z)$ using Numerical Method

Write a code for divergence of given vector function $\vec{A}(x,y,z)$ at the point.

Look at the Jupyter Notebook for this !!



$$\text{Curl } \vec{A} = \vec{\nabla} \times \vec{A} = \hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{j} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\frac{\partial A_z}{\partial y} = \frac{A_z(x, y+h, z) - A_z(x, y-h, z)}{2h}$$

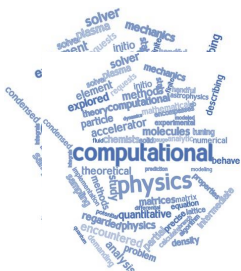
$$\frac{\partial A_y}{\partial z} = \frac{A_y(x,y,z+h) - A_y(x,y,z-h)}{2h}$$

$$\frac{\partial A_x}{\partial z} = \frac{A_x(x,y,z+h) - A_x(x,y,z-h)}{2h}$$

$$\frac{\partial A_z}{\partial x} = \frac{A_z(x+h,y,z) - A_z(x-h,y,z)}{2h}$$

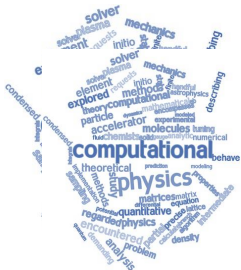
$$\frac{\partial A_y}{\partial x} = \frac{A_y(x+h,y,z) - A_y(x-h,y,z)}{2h}$$

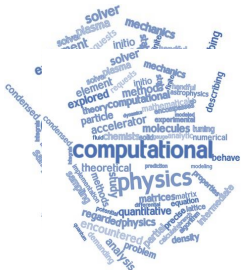
$$\frac{\partial A_x}{\partial y} = \frac{A_x(x, y+h, z) - A_x(x, y-h, z)}{2h}$$



Write a code for determining curl of given vector function $\vec{A}(x,y,z)$ at the point.

Look at the Jupyter Notebook for this !!





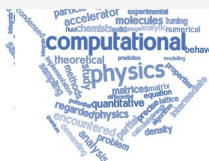
Matrix

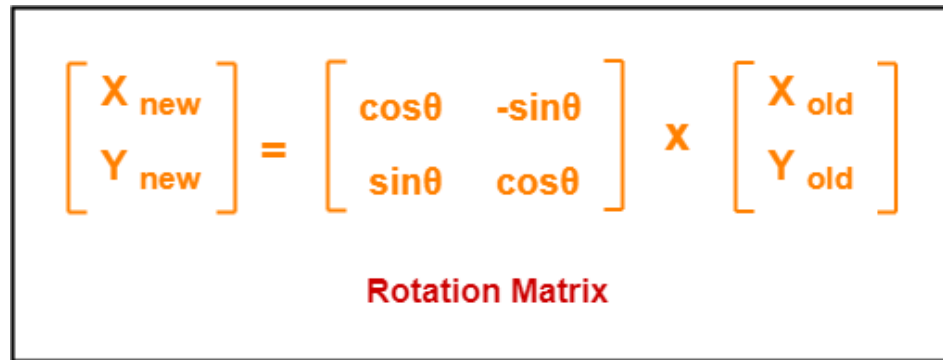
Matrix, a set of numbers arranged in rows and columns so as to form a rectangular array. The numbers are called the elements, or entries, of the matrix



Matrix In Mathematics

$$A = \begin{matrix} & \xrightarrow{\text{Row (m)}} & \\ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} & & \downarrow \text{Columns (n)} \end{matrix}$$



[illegible]



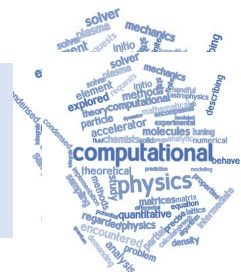
1	5	9	13
2	6	10	14
3	7	11	15
4	8	12	16



Rows reversed of the Transpose matrix.



Rotation of a mobile device from portrait to landscape using matrix algebra



Matrix-Applications

$$2x_1 + x_2 + 2x_3 + x_4 = 6$$

$$6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$$

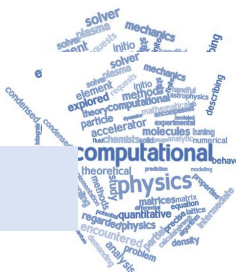
$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

$$\begin{bmatrix} 2 & 1 & 2 & 1 \\ 6 & -6 & 6 & 12 \\ 4 & 3 & 3 & -3 \\ 2 & 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 6 \\ 36 \\ -1 \\ 10 \end{bmatrix}$$

$$AX=H$$

$$X=A^{-1}H$$

Matrix algebra can be used in solving the equations



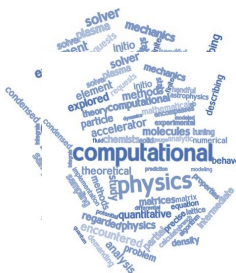
The general form of the Matrix

n is the number of columns

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & a_{24} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & a_{34} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & a_{m4} & \cdots & a_{mn} \end{bmatrix}$$

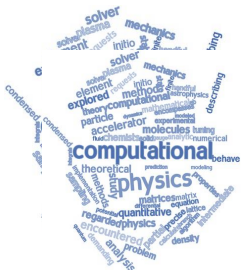
m is the number of rows

$$A = \{a_{ij}\}_{m \times n}$$





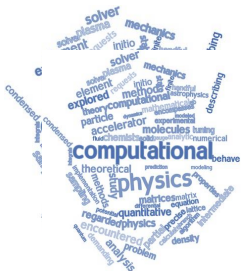
If $m \neq n$



$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

If $m=n$

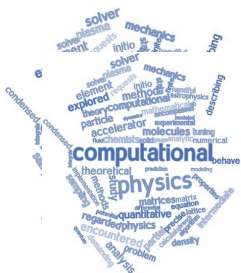
A is Square Matrix

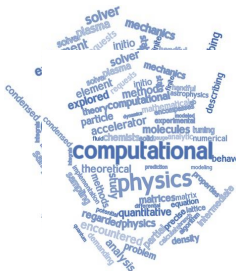


Diagonal Matrix

$$A = \begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ 0 & a_{22} & 0 & 0 & 0 \\ 0 & 0 & a_{33} & 0 & 0 \\ 0 & 0 & 0 & a_{44} & 0 \\ 0 & 0 & 0 & 0 & a_{55} \end{bmatrix}_{5 \times 5}$$

A is a Diagonal Matrix



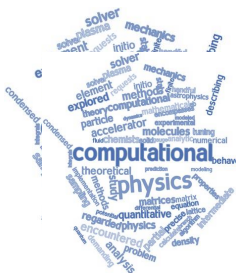
$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$


Type of the Matrix

$$I_n = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix} \text{ Identity Matrix}$$

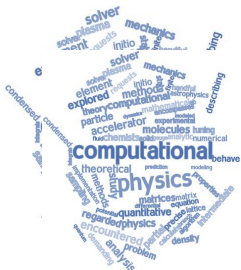
In a square matrix, if all the diagonal elements are 1

Then, I_n is called the identity matrix



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Matrix Operation: Addition -Properties

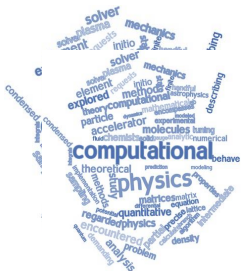
Matrix Operation: Addition -Properties

- ## Matrix Operation: Addition -Properties



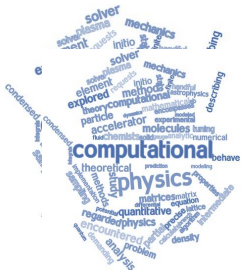
Subtraction of Matrices

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} - \begin{bmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ b_{m1} & b_{m2} & b_{m3} & \dots & b_{mn} \end{bmatrix} = \begin{bmatrix} a_{11} - b_{11} & a_{12} - b_{12} & \dots & a_{1n} - b_{1n} \\ a_{21} - b_{21} & a_{22} - b_{22} & \dots & a_{2n} - b_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} - b_{m1} & a_{m2} - b_{m2} & \dots & a_{mn} - b_{mn} \end{bmatrix}$$





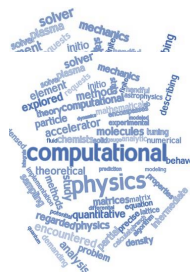
C IS SCALAR



$$= \begin{bmatrix} 1 \times 10 + 2 \times 20 + 3 \times 30 & 1 \times 11 + 2 \times 21 + 3 \times 31 \\ 4 \times 10 + 5 \times 20 + 6 \times 30 & 4 \times 11 + 5 \times 21 + 6 \times 31 \end{bmatrix}$$

$$= \begin{bmatrix} 10+40+90 & 11+42+93 \\ 40+100+180 & 44+105+186 \end{bmatrix} = \begin{bmatrix} 140 & 146 \\ 320 & 335 \end{bmatrix}$$

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} + \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$$



Matrix Operation: Multiplication

$$\text{row } i \begin{bmatrix} \text{---} \end{bmatrix} \times \begin{bmatrix} \text{col } j \\ \text{---} \end{bmatrix} = \begin{bmatrix} c_{ij} \end{bmatrix}$$

$c_{ij} = \text{row } i \times \text{col } j = \sum_{k=1}^n a_{ik} b_{kj}$

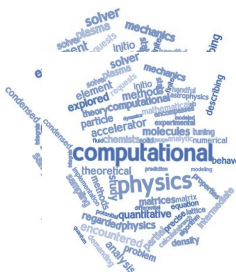
$$A = \{a_{ik}\}_{m \times p}$$

$$B = \{a_{kj}\}_{p \times n}$$

$$AXB = \left\{ \sum_{k=1}^p a_{ik} \times b_{kj} \right\}_{m \times n}$$

$$AXB = \{ab_{ij}\}_{m \times n}$$

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$$

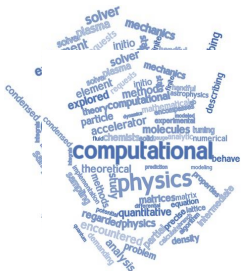


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Matrix Operation: Multiplication-Properties

$$\mathbf{ABC}=\mathbf{D}=\{\mathbf{d}_{ij}\}_{m \times n}$$

- $(AB)C = A(BC)$
- $A(B+C) = AB+AC$
- $(A+B)C = AC+BC$
- $IA = AI$



Question

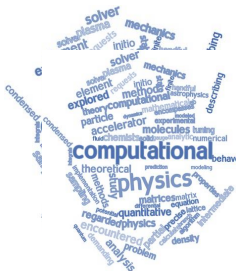
$$A = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ 3 & -1 & 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 2 & -1 \\ -1 & 4 & 2 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & -2 \\ -4 & 1 & 2 \end{pmatrix}$$

Evaluate $A^3 - 4A^2 - 3A$

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{pmatrix}$$



Matrix Operation: Transpose

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{i3} & \cdots & a_{3n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

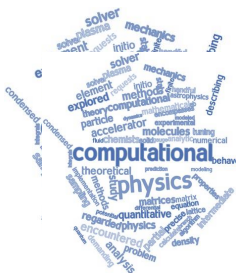
$$A = \{a_{ij}\}_{m \times n}$$



$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} & \cdots & \cdots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \cdots & \cdots & a_{m2} \\ a_{13} & a_{23} & a_{33} & \cdots & \cdots & a_{m3} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & \cdots & a_{mn} \end{bmatrix}$$

$$A^T = \{a_{ji}\}_{n \times m}$$

A^T is called the transpose of matrix A



A	A ^T
$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$
$\begin{bmatrix} 5 \end{bmatrix}$	$\begin{bmatrix} 5 \end{bmatrix}$
$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}$	$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \end{bmatrix}$
$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$	$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$

Matrix Operation: Transpose

$$A = \{a_{ij}\}_{m \times m}$$

$$(A^T) = A$$



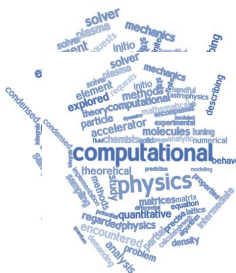
The A is called the symmetry matrix

$$(A^T) = -A$$



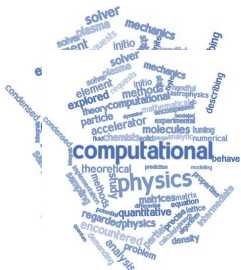
The A is called the skew symmetry matrix

Symmetric Matrices	Skew Symmetric Matrices
$\begin{bmatrix} x & a & b \\ a & y & c \\ b & c & z \end{bmatrix}$	$\begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$



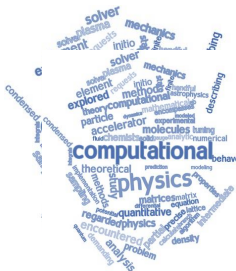
$$\mathbf{k}(\mathbf{A}^T) = (\mathbf{k}\mathbf{A})^T$$

$$(\mathbf{AB})^T = (\mathbf{B}^T \mathbf{A}^T)$$



Matrix Operation: Transpose Properties

Prove that any square matrix can be expressed as the sum of a symmetric matrix and a skew-symmetric matrix.



Matrix Operation: Transpose

If A is a square matrix, $(A + A^T)$ is symmetric and $(A - A^T)$ is skew-symmetric.

$$B = (A + A^T), \text{ then } B^T = (A + A^T)^T$$

$$B^T = A^T + (A^T)^T = (A^T + A) = (A + A^T) = B$$

B is a Symmetric Matrix

$$C = (A - A^T) \text{ then } C^T = (A - A^T)^T$$

$$C^T = A^T - (A^T)^T = (A^T - A) = -(A - A^T) = -C$$

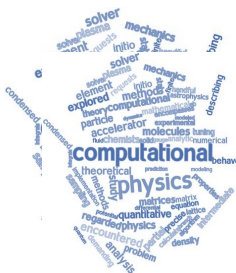
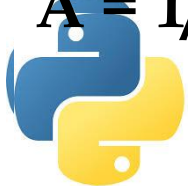
C is Skew Symmetric Matrix

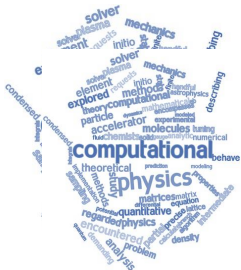
Thus, any square matrix can be expressed as the sum of a symmetric and a skew-symmetric matrix.

$$A = 1/2(B + C) = 1/2 (A + A^T) + 1/2 (A - A^T)$$

Symmetric

Skew symmetric

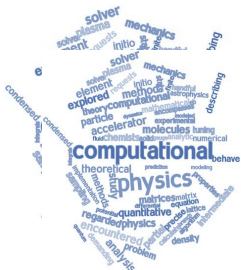




Identity Matrix

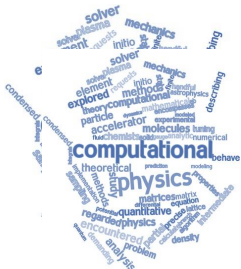
$$\mathbf{I}A=A\mathbf{I}=A$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$



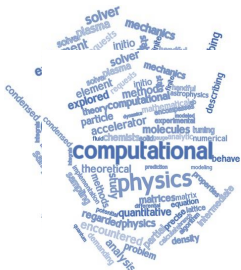
Write a code for matrix scalar-Matrix Multiplication, Matrix addition and subtraction using 2d array

Look at the Jupyter Notebook for this !!



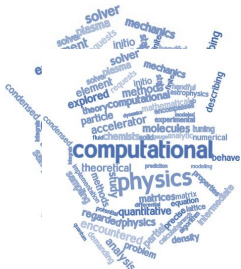
Write a code for matrix multiplication using for loop and 2d array

Look at the Jupyter Notebook for this !!



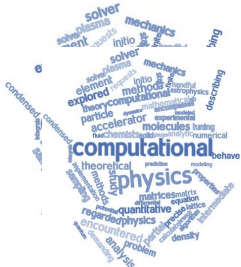
**Write a code for matrix scalar-matrix
Multiplication, Matrix multiplication, addition
and subtraction using matrix function using the
NumPy Library**

Look at the Jupyter Notebook for this !!



Write a code for Transpose of Matrix using for loop and 2d array

Look at the Jupyter Notebook for this !!



Matrix Operation: Inverse of Matrix A

If A is a square matrix of order $m \times m$, and if there exists another square matrix B of the same order $m \times m$, such that $AB = BA = I$, then, A is said to be an **invertible matrix** and B is called the **inverse matrix of A** and it is denoted by A^{-1} .

$$BA=AB=I$$

Then Matrix B will be called the inverse of Matrix A:

$$A^{-1}=B$$

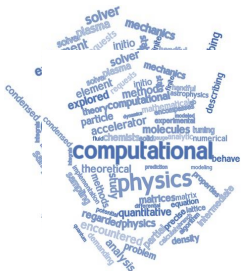


$$A^{-1}A=I = AA^{-1}$$

A is invertible (or nonsingular) i.e. $\det(A) \neq 0$ if there exists another matrix A^{-1} such that:

- 

$$(\mathbf{A}^{-1})^T = (\mathbf{A}^T)^{-1}$$



The elementary row operations for the transformation of the matrix

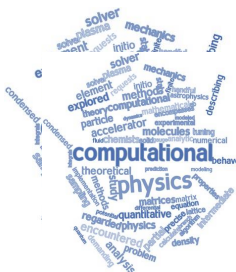
The three basic elementary operations or transformation of a matrix are:

(1) The interchange of any two rows

$$R_2 \leftrightarrow R_3 \quad \begin{bmatrix} 1 & -1 & 2 & 0 \\ -2 & 3 & 7 & 1 \\ 4 & 5 & -6 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 & 0 \\ 4 & 5 & -6 & -3 \\ -2 & 3 & 7 & 1 \end{bmatrix}$$

(2) The multiplication of the elements of any row by a non zero number

$$\begin{bmatrix} 1 & 2 & 3 & | & 4 \\ 5 & 6 & 7 & | & 8 \\ 9 & 10 & 11 & | & 12 \end{bmatrix} \xrightarrow{(3)} \begin{bmatrix} 1 & 2 & 3 & | & 4 \\ 15 & 18 & 21 & | & 24 \\ 9 & 10 & 11 & | & 12 \end{bmatrix}$$

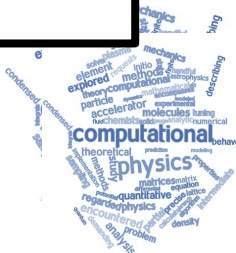


The elementary operations for the transformation of the matrix

(3) The addition to the elements of any row, the corresponding elements of any other row multiplied by any non-zero number.

$$\begin{bmatrix} 1 & 2 & 3 & | & 4 \\ 5 & 6 & 7 & | & 8 \\ 9 & 10 & 11 & | & 12 \end{bmatrix} \xrightarrow{-5R_1} \begin{bmatrix} 1 & 2 & 3 & | & 4 \\ 5-5(1) & 6-5(2) & 7-5(3) & | & 8-5(4) \\ 9 & 10 & 11 & | & 12 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & | & 4 \\ 0 & -4 & -8 & | & -12 \\ 9 & 10 & 11 & | & 12 \end{bmatrix}$$

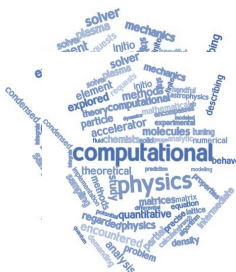


Inverse of a Matrix using Elementary Row Operations

This is a fun way to find the Inverse of a Matrix:

$$\left[\begin{array}{c|c} A & I \end{array} \right]$$
"Elementary Row Operations"

$$\left[\begin{array}{c|c} I & A^{-1} \end{array} \right]$$



Method

Augmented matrix

$$[A \mid I] \xrightarrow{\text{row operations}} [I \mid A^{-1}]$$

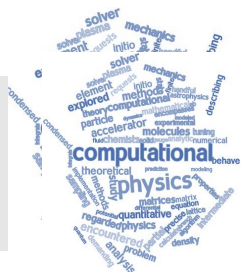
A^{-1} is the inverse of the matrix of A obtained after applying the elementary row operations to A .

Look at this example to understand !!



[1 | 1/8]

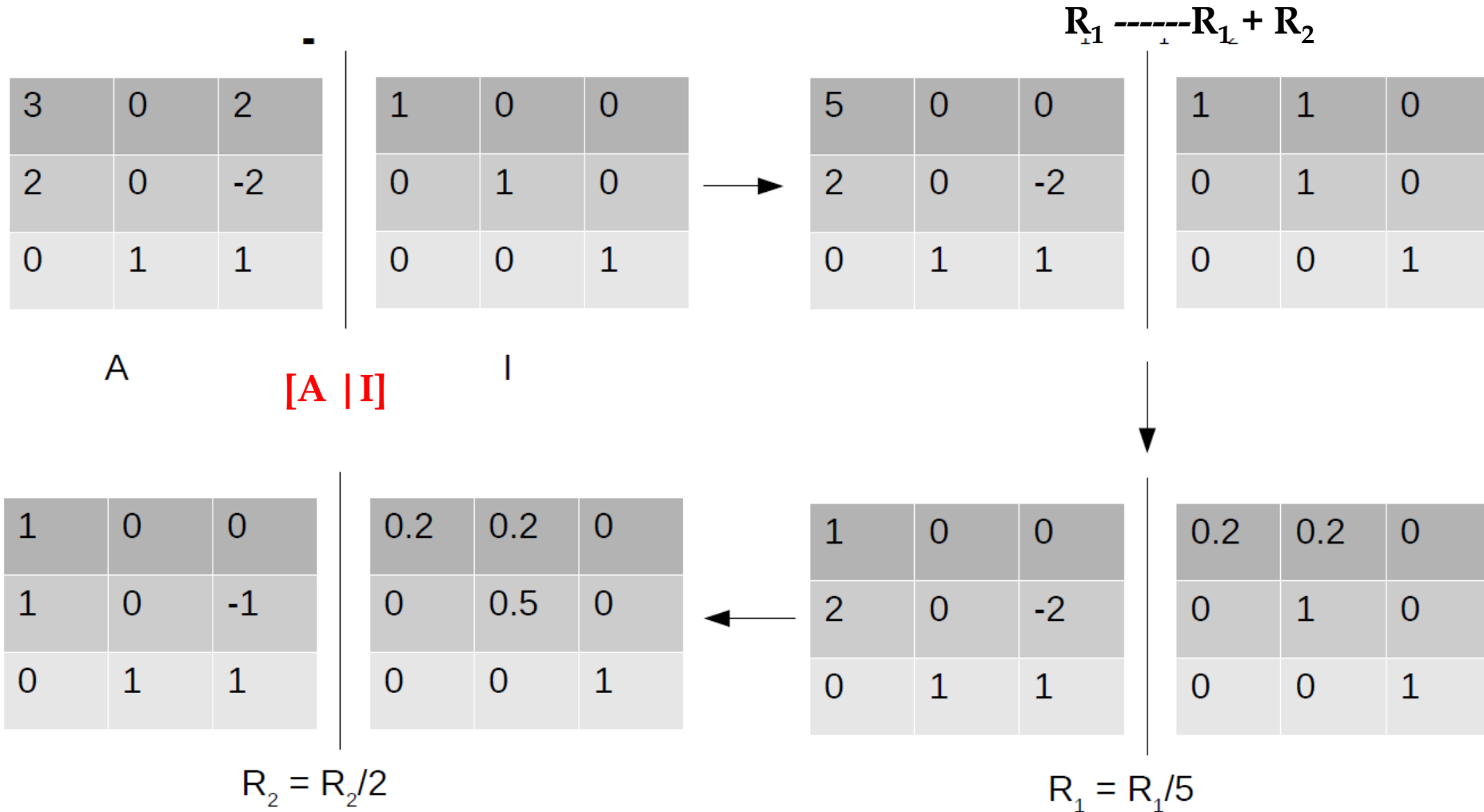
$1/8$ is the multiplicative inverse of 8, obtained by dividing 1 by 8. The number 1 acts as the identity element in multiplication, similar to how the identity matrix is used in augmented form



Inverse of Matrix A using the transformation method

Row Transformation

$$[A \mid I] \xrightarrow{\quad\quad\quad} [I \mid A^{-1}]$$



Inverse of Matrix A using the transformation method

$$\begin{array}{ccc|ccc}
 1 & 0 & 0 & 0.2 & 0.2 & 0 \\
 0 & 0 & -1 & -0.2 & 0.3 & 0 \\
 0 & 1 & 1 & 0 & 0 & 1
 \end{array}
 \xrightarrow{R_2 = R_2 - R_1}
 \begin{array}{ccc|ccc}
 1 & 0 & 0 & 0.2 & 0.2 & 0 \\
 0 & 1 & 0 & -0.2 & 0.3 & 1 \\
 0 & 1 & 1 & 0 & 0 & 1
 \end{array}
 \xrightarrow{R_2 = R_2 + R_3}
 \begin{array}{ccc|ccc}
 1 & 0 & 0 & 0.2 & 0.2 & 0 \\
 0 & 1 & 0 & -0.2 & 0.3 & 1 \\
 0 & 0 & 1 & 0 & 0 & 1
 \end{array}$$

The elementary row operations

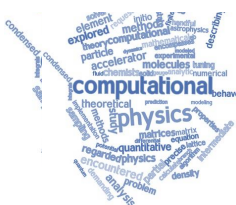
$$[A \mid I] \xrightarrow{\text{elementary row operations}} [I \mid A^{-1}]$$

$$\begin{array}{ccc|ccc}
 1 & 0 & 0 & 0.2 & 0.2 & 0 \\
 0 & 1 & 0 & -0.2 & 0.3 & 1 \\
 0 & 0 & 1 & 0.2 & -0.3 & 0
 \end{array}
 \xrightarrow{R_3 = R_3 - R_2}
 \begin{array}{ccc|ccc}
 1 & 0 & 0 & 0.2 & 0.2 & 0 \\
 0 & 1 & 0 & -0.2 & 0.3 & 1 \\
 0 & 0 & 1 & 0.2 & -0.3 & 0
 \end{array}$$

$[I \mid A^{-1}]$

<https://www.mathsisfun.com/algebra/matrix-inverse-row-operations-gauss-jordan.html>

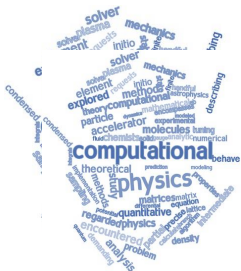
<https://www.youtube.com/watch?v=bYrSXv8IM5Y>



Question

Find the inverse of the given matrix

3	2	1
2	3	1
1	1	4



[illegible]

1	0	0		0.611	-0.389	-0.056
0	1	0		-0.389	0.611	-0.055
0	0	1		-0.056	-0.055	0.278

$$[\mathbf{I} \mid \mathbf{A}^{-1}]$$

Method

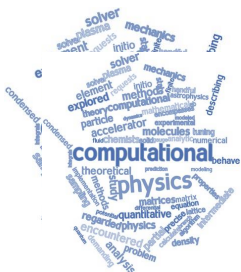
[A | I]

3	2	1		1	0	0
2	3	1		0	1	0
1	1	4		0	0	1

Steps



1. Upper triangular matrix through row operations
2. Lower triangular matrix through row operations
3. Make diagonal element values 1 through scalar row operations



[illegible]

[illegible]

e

3	2	1		1	0	0
0	1.667	0.333		-0.667	1	0
0	2	4		-1	1	1

e

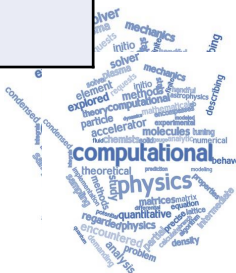
3	2	1		1	0	0
0	1.667	0.333		-0.667	1	0
0	0	3.6		-0.2	-0.2	1

Inverse of Matrix A using the transformation method

3	2	1		1	0	0
0	1.667	0.333		-0.667	1	0
0	0	3.6		-0.2	-0.2	1

$$R_1 \leftarrow R_1 - (R_3)/3.6$$

3	2	0		1.056	0.055	-0.278
0	1.667	0.333		-0.667	1	0
0	0	3.6		-0.2	-0.2	1



[illegible]

3	2	0		1.056	0.055	-0.278
0.	1.667	0		-0.648	1.018	-0.093
0	0	3.6		-0.2	-0.2	1

3	0	0		1.833	-1.166	-0.167
0.	1.667	0		-0.648	1.018	-0.093
0	0	3.6		-0.2	-0.2	1

Inverse of Matrix A using the transformation method

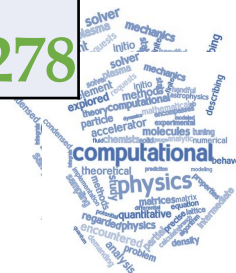
3	0	0		1.833	-1.166	-0.167
0.	1.667	0		-0.648	1.018	-0.093
0	0	3.6		-0.2	-0.2	1

$$R_1 \text{-----} R_1/3$$

$$R_2 \text{----} R_2/1.667$$

$R_3 \text{----} R_3/3.6$

1	0	0		0.611	-0.389	-0.056
0	1	0		-0.389	0.611	-0.055
0	0	1		-0.056	-0.055	0.278



Inverse of Matrix A using the transformation method

Input:

A square matrix A of size $n \times n$.

Output:

Inverse of matrix A (if it exists).

Steps:

1. Check if the matrix is square:

- 1. Ensure the matrix A is square. The number of rows should be equal to the number of columns.**

2. Calculate the determinant:

1. Compute the determinant of matrix A.
2. If the determinant is 0, the matrix is singular, and no inverse exists. Exit the algorithm.

3. Create the augmented matrix:

1. Construct an augmented matrix by placing the identity matrix I of the same size next to A . This forms a matrix $[A|I]$.

4. Perform Gaussian elimination:

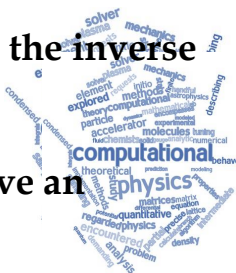
1. Use row operations to transform the matrix A on the left into the identity matrix III.
2. The row operations include:
 1. Swapping rows.
 2. Multiplying a row by a non-zero scalar.
 3. Adding or subtracting multiples of rows from each other.
3. Apply the same row operations to the identity matrix III on the right.
4. Once the left side becomes the identity matrix, the right side will be transformed into A^{-1} , the inverse of A.

5. Return the inverse:

1. If the left side of the augmented matrix is now the identity matrix, the right side will be the inverse of A.

6. Check for errors:

1. If the matrix is not invertible, return an error message indicating the matrix does not have an inverse.



Adjoint and Determinant method to find the Inverse of Matrix A

Inverse of the matrix can also be calculated using $A^{-1} = \frac{\text{adj}(A)}{|A|}$

Consider a 3x3 matrix

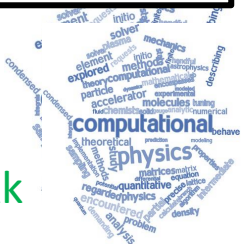
$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

Its cofactor matrix is

$$C = \begin{pmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{pmatrix} \quad \text{adj}(A) = C^T = \begin{pmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \\ + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{pmatrix}$$

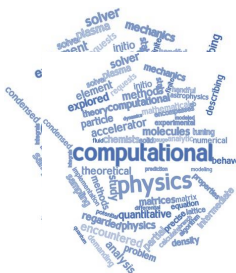


<https://www.youtube.com/watch?v=lauh7KjEfXk>



$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \\ \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \\ \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 \end{bmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$$



Which method is more efficient for the computer

Transformation method

Adjoint and Determinant method

?

Cofactor Expansion Complexity: finding the adjoint of a matrix requires computing the cofactors for each element of the matrix. This involves calculating multiple determinants of submatrices, which quickly becomes computationally expensive as the size of the matrix increases.

Numerical Instability: The adjoint and determinant method is more prone to numerical instability. Small rounding errors in computing the determinant and cofactors can lead to large inaccuracies in the final inverse, especially for larger matrices.

Finding solutions of equations using Matrix method

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = h_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = h_2$$

.

.

.

.

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n = h_n$$

$$AX = H$$

$$A^{-1}AX = A^{-1}H$$

$$IX = A^{-1}H$$

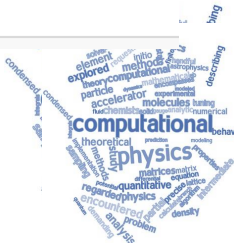
$$X = A^{-1}H$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ . & . & . & . & . \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{bmatrix}$$

$$H = \begin{bmatrix} h_1 \\ h_2 \\ . \\ . \\ h_n \end{bmatrix}$$

$$AX=H$$



AX=H

$$\mathbf{A}^{-1}\mathbf{A}\mathbf{X}=\mathbf{A}^{-1}\mathbf{H}$$



$$\mathbf{X} = \mathbf{A}^{-1} \mathbf{H}$$

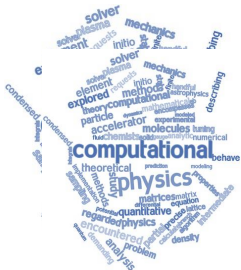
Determine A^{-1} using the transformation method.

Then, compute the matrix product of A^{-1} with the matrix H to obtain X for solutions of equations.



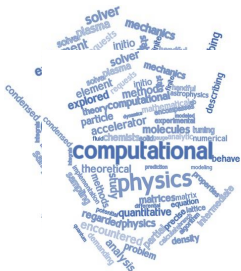
$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

Find the solutions through the Matrix method ?



$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

1. $AX=H$
2. $X=A^{-1}H$
3. Find A^{-1} using the transformation method
4. $X=A^{-1}H$



Question- Hint

$$2x_1 + x_2 + 2x_3 + x_4 = 6$$

$$6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$$

$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

$$A = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 6 & -6 & 6 & 12 \\ 4 & 3 & 3 & -3 \\ 2 & 2 & -1 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

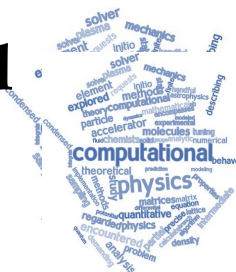
$$H = \begin{bmatrix} 6 \\ 36 \\ -1 \\ 10 \end{bmatrix}$$

1. $AX=H$

2. $X=A^{-1}H$

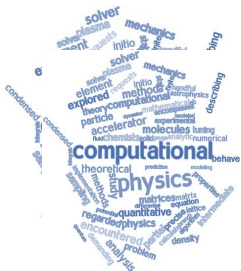
3. Find A^{-1} using the transformation method

4. $X=A^{-1}H$



$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

- 



Question- Hint

$$2x_1 + x_2 + 2x_3 + x_4 = 6$$

$$6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$$

$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

$$A = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 6 & -6 & 6 & 12 \\ 4 & 3 & 3 & -3 \\ 2 & 2 & -1 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

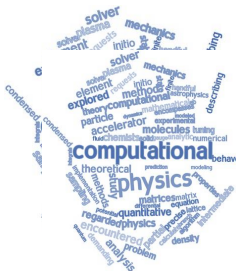
$$H = \begin{bmatrix} 6 \\ 36 \\ -1 \\ 10 \end{bmatrix}$$

1. $AX=H$

2. $X=A^{-1}H$

3. Find $A^{-1} = \text{Adj}(A) / |A|$

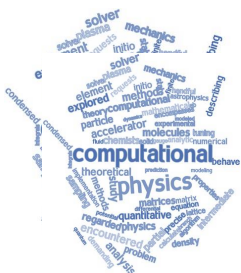
4. $X=A^{-1}H$



$$2\mathbf{x}_1 + 2\mathbf{x}_2 - \mathbf{x}_3 + \mathbf{x}_4 = 10$$

Solve it using the **Gauss-Jordan** method to find X

[https://math.libretexts.org/Bookshelves/Applied_Mathematics/Applied_Finite_Mathematics_\(Sekhon_and_Bloom\)/02%3A_Matrices/2.02%3A_Systems_of_Linear_Equations_and_the_Gauss-Jordan_Method](https://math.libretexts.org/Bookshelves/Applied_Mathematics/Applied_Finite_Mathematics_(Sekhon_and_Bloom)/02%3A_Matrices/2.02%3A_Systems_of_Linear_Equations_and_the_Gauss-Jordan_Method)



Hint- An alternate way to solve this is by using the transformation matrix.

$$2x_1 + x_2 + 2x_3 + x_4 = 6$$

$$6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$$

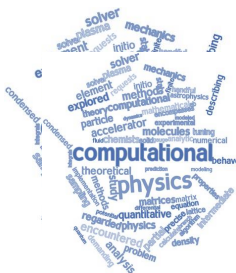
$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$$

$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

$$A = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 6 & -6 & 6 & 12 \\ 4 & 3 & 3 & -3 \\ 2 & 2 & -1 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}$$

$$H = \begin{bmatrix} 6 \\ 36 \\ -1 \\ 10 \end{bmatrix}$$



Question- Hint

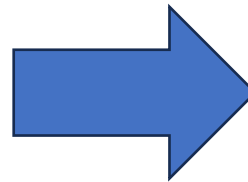
$$A = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 6 & -6 & 6 & 12 \\ 4 & 3 & 3 & -3 \\ 2 & 2 & -1 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$H = \begin{bmatrix} 6 \\ 36 \\ -1 \\ 10 \end{bmatrix}$$

Apply row operations

$$\left[\begin{array}{cccc|c} 2 & 1 & 2 & 1 & 6 \\ 6 & -6 & 6 & 12 & 36 \\ 4 & 3 & 3 & -3 & -1 \\ 2 & 2 & -1 & 1 & 10 \end{array} \right]$$



$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & c_1 \\ 0 & 1 & 0 & 0 & c_2 \\ 0 & 0 & 1 & 0 & c_3 \\ 0 & 0 & 0 & 1 & c_4 \end{array} \right]$$

Augmented matrix

$$[A \mid H] \text{-----} [I \mid C]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$



<https://www.youtube.com/watch?v=CsTOUbeMPUo>

To find eigen values first we find the characteristic equation given by

$$| A - \lambda I | = 0$$

Now the roots of the equation or values of λ are called eigen values.

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Eigenvalues and Eigen vectors of Matrix A

$$\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Taking $\lambda = -2$

$$\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -2 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$0 \cdot x_1 + x_2 = -2 \cdot x_1 \quad x_2 = -2 \cdot x_1 \quad (1)$$

$$-2 \cdot x_1 - 3 \cdot x_2 = -2 \cdot x_2 \quad -x_2 = 2 \cdot x_1 \quad (2)$$

Taking $x_1 = 1$ then $x_2 = -2$

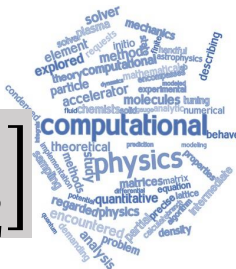
$$X = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Eigen vectors of $\lambda = -1$

$$X = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

the Eigen vectors of $\lambda = 2$

$$X = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$



EXAMPLE

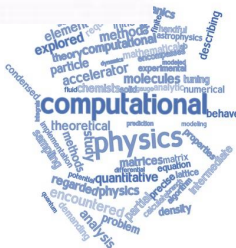
Let A be a square matrix of order 3×3

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

According to Cayley Hamilton theorem

$$|A - \lambda I| = 0$$

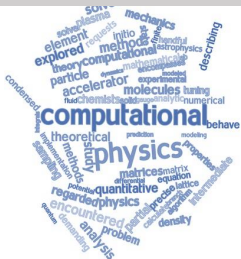
Where λ is any scalar, and
I is the 3×3 unit matrix



$$\begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} = 0$$
$$\lambda^3 - 7\lambda^2 + 36 = 0$$
$$\lambda = -2, 3, 6$$

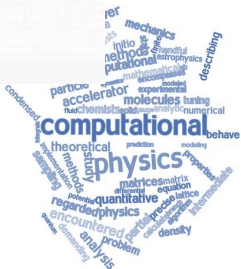
These are called eigen values.

Find the Eigen Vectors for eigen values $\lambda=-2, 3$ and 6



PROPERTIES OF EIGEN VALUES

1. Any square matrix A and its transpose A' have the same eigen values.
2. The sum of the eigen values of a matrix is equal to the trace of the matrix.
3. The product of the eigen values of a matrix A is equal to the determinant of A .
4. The eigen values of a triangular matrix are just the diagonal elements of the matrix.
5. The eigen values of an idempotent matrix are either zero or unity.



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